

Name: _____



New York State Testing Program

Elementary-level Science Test

Grade 5

Spring 2026

RELEASED QUESTIONS

Elementary-level Science Test



TIPS FOR TAKING THE TEST

Here are some ideas to help you do your best:

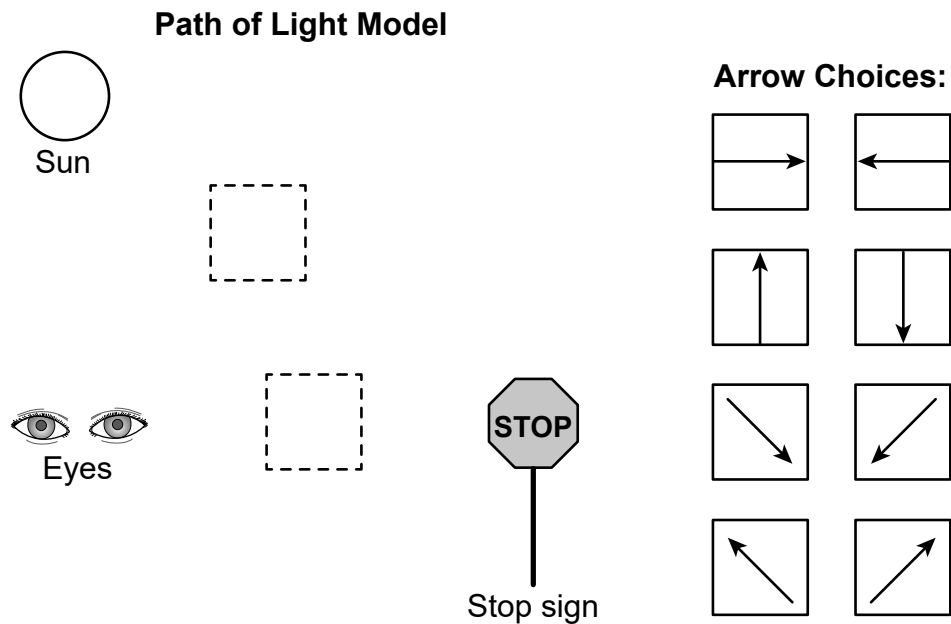
- Be sure to read all the directions carefully.
- Read each question carefully.
- Think about the answer before making your choice or writing your answer.
- Make sure you read all the information given for each question.
- Diagrams included in the test are not drawn to scale unless otherwise noted.
- You have a ruler and a calculator that you can use on the test if they help you answer the question.

Base your answers to questions 1 through 4 on the information below and on your knowledge of science.

Sunlight and Starlight

Our solar system exists in the Milky Way Galaxy. There are about 100 billion stars in the Milky Way and the Sun is one of them. The Sun is the only star in our solar system. The Sun is responsible for many patterns and events that occur on Earth.

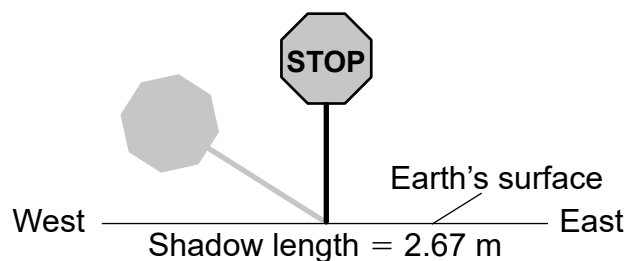
- 1
- Sunlight allows objects to be visible on Earth. From the choices, place one arrow in *each* box to complete the model to show the correct path of light that permits a stop sign to be seen. [1]



The data below shows some information about the shadow formed by sunlight for a 2.0-meter tall stop sign in Rochester, New York on September 23, 2023.

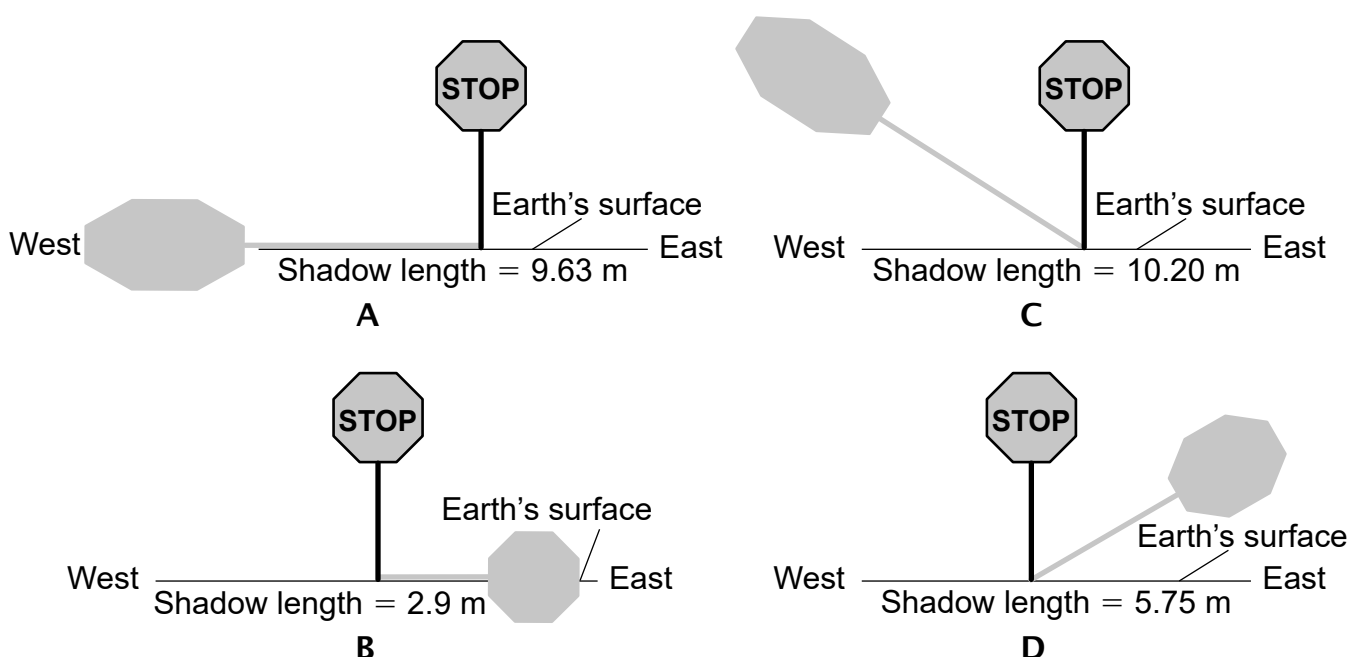
Time of Day	Length of Stop Sign Shadow (m)
8:15 a.m.	8.63
10:45 a.m.	2.67
1:15 p.m.	1.89
3:45 p.m.	3.02
6:15 p.m.	13.21
8:45 p.m.	None

The direction and length of the shadow of the 2.0-meter stop sign for 10:45 a.m. are represented in the diagram below.



2

Which diagram best represents the direction and length of the same stop sign's shadow at 5:15 p.m.?



The data table below shows some information about sunrise and sunset in Rochester, New York on September 23, 2023.

	Time of Day
Sunrise	6:58 a.m.
Sunset	7:07 p.m.

3

Which table correctly identifies the amount of daylight (hours and minutes) in Rochester, New York on December 21, 2023 and June 20, 2024?

A

Month	Daylight
December	8 h 59 min
June	15 h 24 min

B

Month	Daylight
December	15 h 24 min
June	8 h 59 min

C

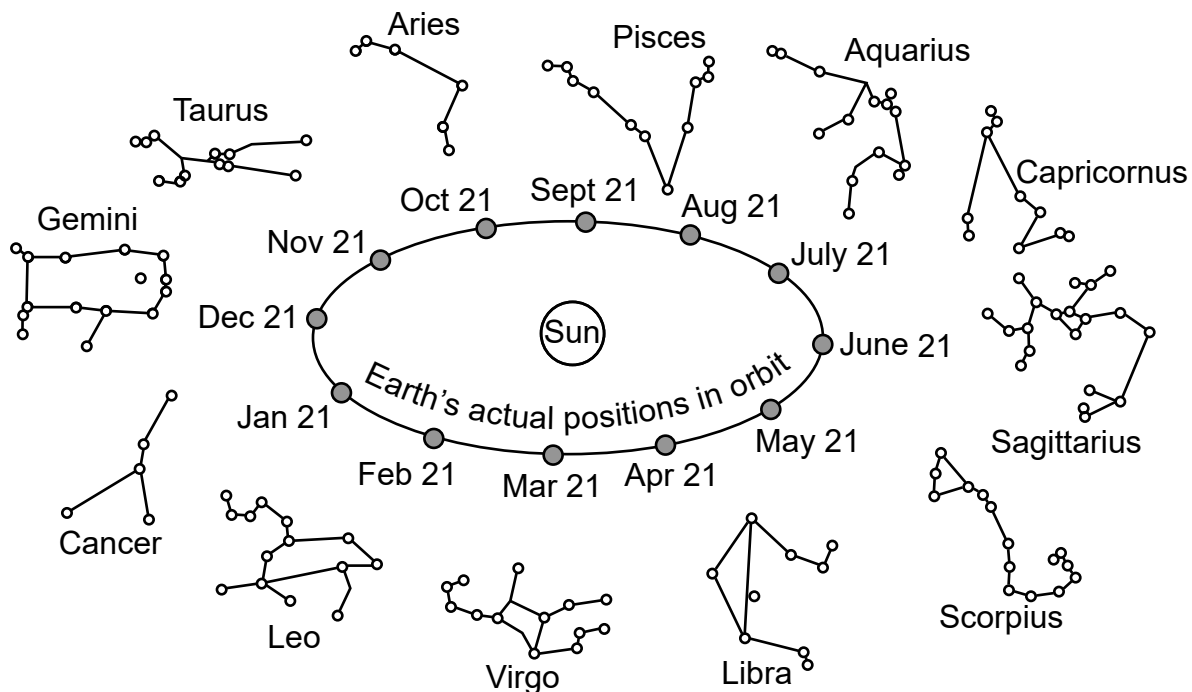
Month	Daylight
December	12 h 9 min
June	11 h 51 min

D

Month	Daylight
December	11 h 51 min
June	12 h 9 min

After the Sun sets, some stars are visible in the night sky. The model below shows different groups of stars (constellations) that are visible in the night sky in Rochester, New York as Earth orbits the Sun.

Model of Constellations



4

From the choices below, place **one** correct constellation name in *each* box to indicate the constellations that would be entirely visible and would **not** be visible from Rochester, New York at night on March 21. [1]

Choices for Constellations:

Gemini

Libra

Pisces

Sagittarius

Virgo

Visible

Not Visible

Base your answers to questions 5 through 9 on the information below and on your knowledge of science.

Minerals of New York State

A variety of minerals are found in New York State. Many minerals are mined and taken out of rock formations. They are used in industry, construction, computers, medicine, and jewelry.

During a unit of study, a teacher gave students five different mineral samples that can be found in New York State. The students used appropriate safety equipment and followed safety procedures as they made observations and performed tests on these minerals to determine their properties. Their observations are shown in the table below.

Minerals of New York State and Their Observed Properties

Mineral	Color	Streak (color in powdered form)	Hardness	Surface Reflectivity	Crystal Shape
calcite	clear/many colors	clear/white	3	non-metallic	cubic
garnet	red-brown	clear	7	non-metallic	cubic
halite	clear/white/ gray	clear/white	2-2.5	non-metallic	cubic
hematite	red-brown	red-brown	5-6.5	non-metallic	hexagonal
talc	pearly white	white	1	non-metallic	not observable

- 5 Which mineral observations could *both* be used to determine that a sample of talc is different from calcite or halite?
- A hardness and crystal shape
 - B streak and crystal shape
 - C hardness and surface reflectivity
 - D streak and surface reflectivity

6

The model represents how three properties can be used to classify these minerals. The areas of overlap show shared properties. Based on the properties shown in *Minerals of New York State and Their Observed Properties*, place the names of **four** of the minerals from the choices below in the correct boxes within the model. [1]

Minerals:

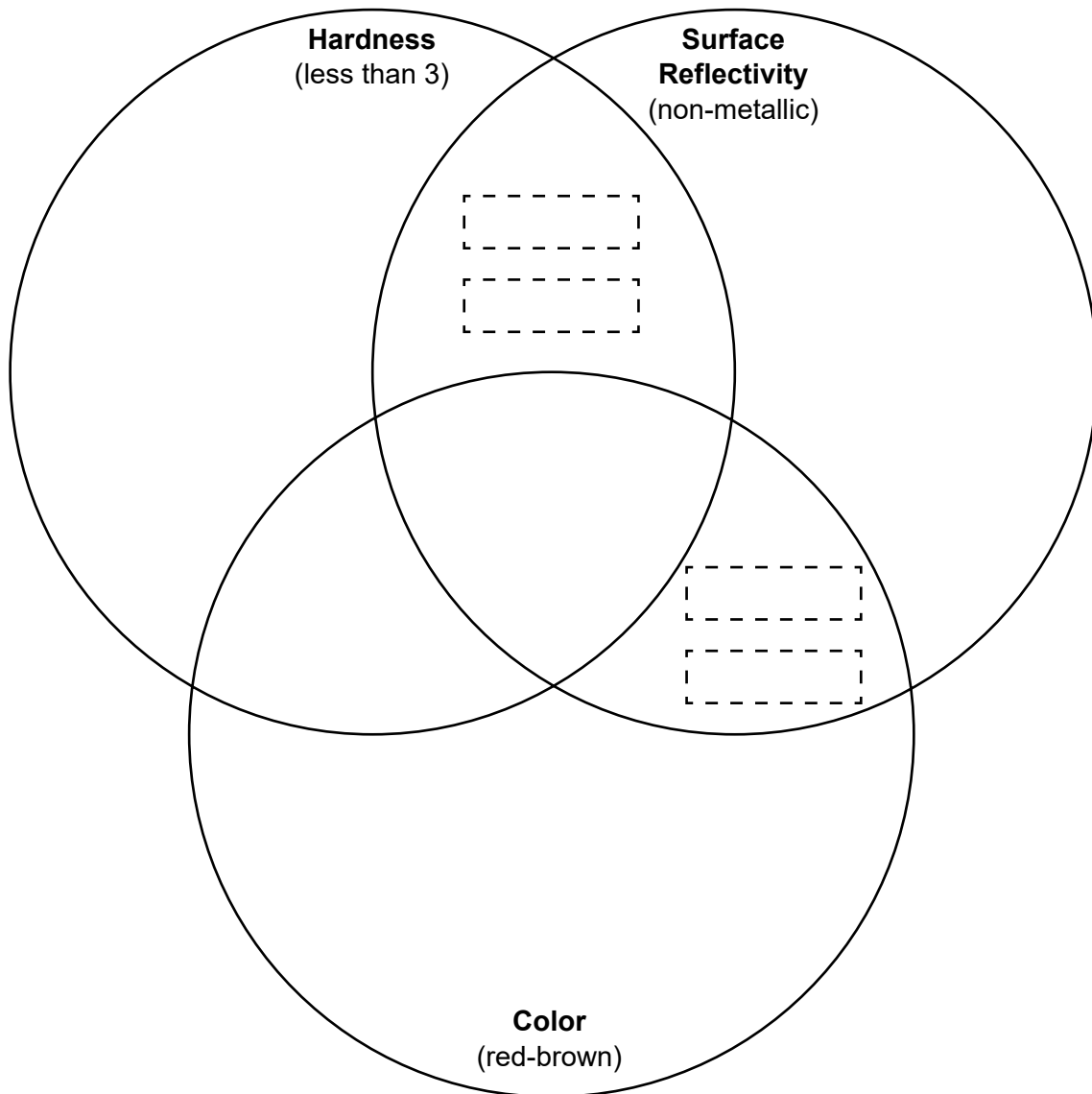
Hematite

Garnet

Talc

Halite

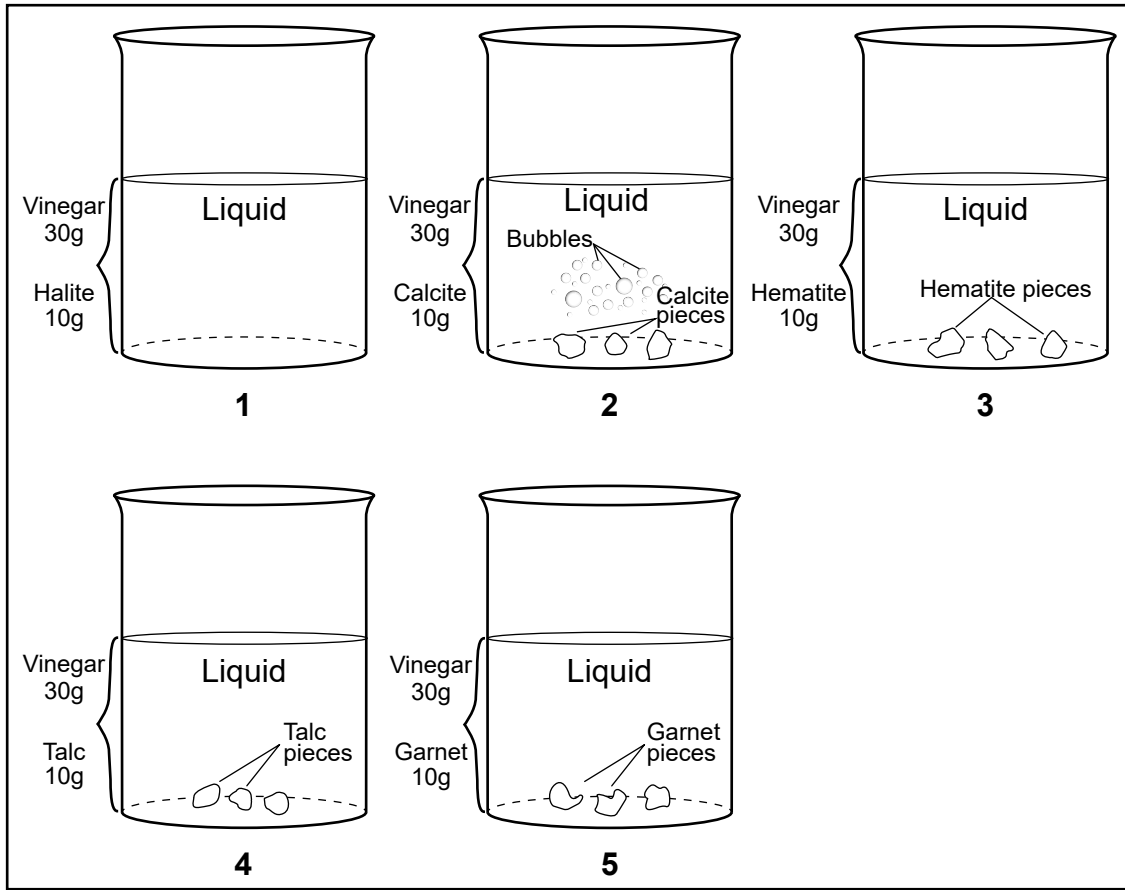
Calcite



The students then investigated how vinegar affects these minerals. They placed equal amounts of each mineral into beakers with equal amounts of vinegar.

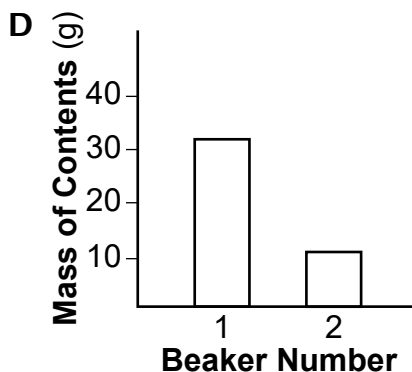
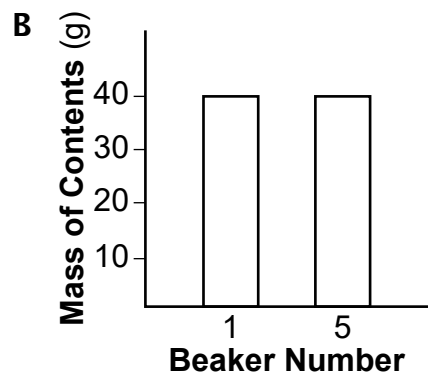
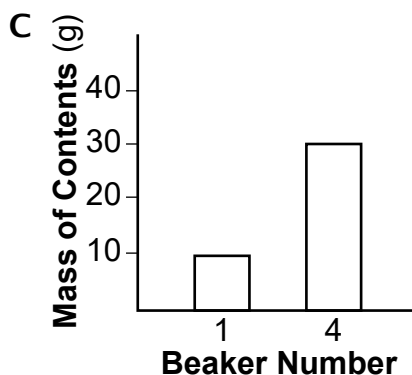
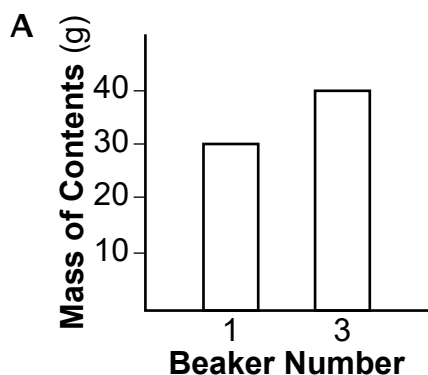
The students made a model, shown below, of the results of the investigation.

Model of Investigation Results



7

Which graph correctly shows the total mass of the contents of two of the beakers in the *Model of Investigation Results*?



8

A student claimed that one of the beakers showed evidence that the mixing of substances resulted in a new substance. Identify the beaker number in which a new substance was formed. Then identify the evidence in the model that supports your choice. [1]

Beaker number: _____

Evidence in model: _____

9

Which piece of evidence correctly identifies an example of solubility?

- A The halite is not visible in the vinegar.
- B The calcite pieces settled to the bottom of the beaker.
- C The hematite pieces became more rounded.
- D The garnet pieces had an increase in reflectivity.

Base your answers to questions 10 through 13 on the information below and on your knowledge of science.

Wolves in Yellowstone National Park

In the 1800s, wolves were commonly found from the cold Arctic regions to Mexico. By the early 1900s, almost no wolves were found in the United States because of habitat loss and hunting by humans.

In 1973, the northern Rocky Mountain wolf was listed as an endangered species. As a solution, wolves were reintroduced into the Greater Yellowstone Ecosystem. From 1995 to 1997, 41 wild wolves from other areas were released into Yellowstone National Park.

Wolves are highly social animals. They eat a wide variety of prey, both large and small. They mate in February, and females give birth to an average of five pups in April.

At the end of 2022, there were 108 wolves in Yellowstone. These wolves lived in 10 different packs, ranging in size from 4 to 25 wolves.

10

Which fact does ***not*** support the argument that being part of a wolf pack helps members survive?

- A** The leading cause of death for wolves outside Yellowstone Park is humans.
- B** Food and protection is provided to newborn pups.
- C** Large prey are hunted and killed by wolves.
- D** Regions in which wolves live are defended from other wolves.

In Yellowstone, about half of the wolves are black in color, and the other half are gray in color.

Studies of these wolves have shown the following:

- Gray-colored wolves are more aggressive than black-colored wolves during encounters with other wolf packs.
- The litter sizes of gray-colored wolves are slightly larger than the litter sizes of black-colored wolves.
- Wolves often choose mates of the opposite color.
- Black-colored wolves have a greater survival rate from some diseases than gray-colored wolves.

Gray-colored and Black-colored Wolves in Yellowstone



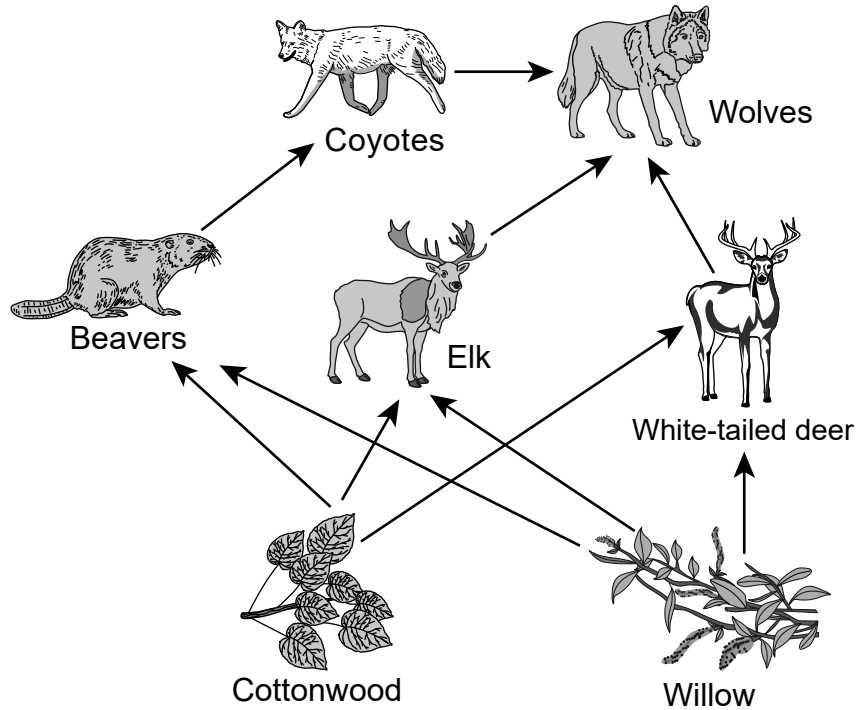
11

Which statement correctly summarizes what the studies found comparing wolves with different coat colors?

- A Wolves have traits that can be influenced by the environment.
- B Variations in characteristics of wolves may provide advantages in surviving and reproducing.
- C Traits of wolves are inherited from parents, and variations of these traits exist in a pack.
- D Internal structures of wolves allow for survival, growth, and successful reproduction.

The incomplete food web model below shows some information about feeding relationships in Yellowstone National Park.

Yellowstone National Park Food Web



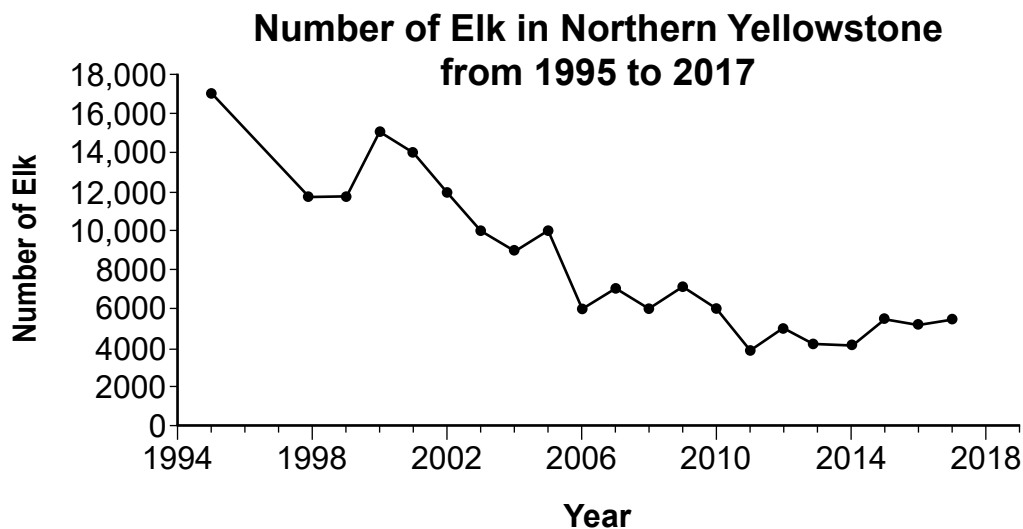
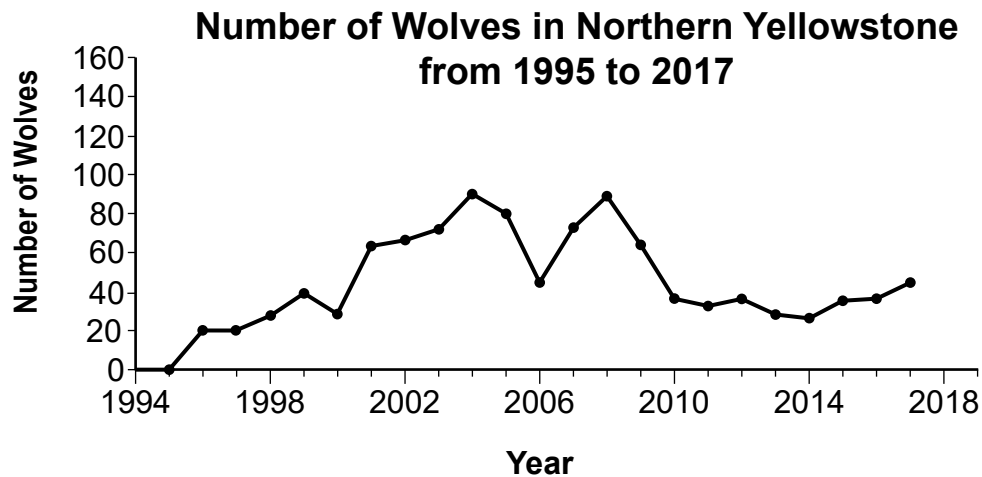
12

Identify **one** food source for the white-tailed deer in Yellowstone National Park. Then, describe how this food source would be affected if the white-tailed deer population changed when more wolves were introduced into the park. Include a reason for this change in your description. [1]

Food source: _____

Effect on food source: _____

The graphs below show some information about the number of wolves and elk found in the northern range of Yellowstone National Park since wolves were reintroduced to the park in 1995.



13

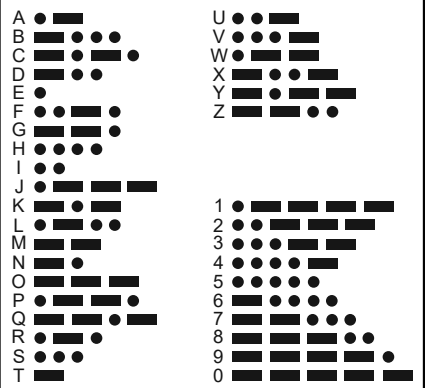
Which piece of evidence from the graphs supports the claim that wolves are dependent on elk for their survival in Yellowstone National Park?

- A From 2010 to 2017, both the wolf and elk populations decreased, then increased.
- B Both the wolf and elk populations increased from 2005 to 2006.
- C From 2000 to 2004, the number of wolves increased while the number of elk decreased.
- D The highest wolf populations were in 2004 and 2008, while the highest elk populations were in 1995 and 2000.

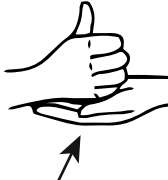
Base your answers to questions 14 through 18 on the information below and on your knowledge of science.

Information Transfer

People communicate in order to share information with others, express needs, and maintain friendships. Throughout history, people around the world have created and used different methods of communication. The table below describes a few of these methods.

Method	Description	Example
Drum signals	System that uses different sound combinations on a drum.	 Mallet Drum
Morse code	System that uses long and short sounds, long or short flashing lights, or dots and dashes, to represent letters and numbers.	
String-and-can telephone	System that allows a person to talk into a can on one end and be heard through the can on the other end.	

(Table continued)

Method	Description	Example																																																																				
Sign language	System that uses hand movements to represent letters and words.	Help  Finished  More  Play 																																																																				
Binary code	System that uses 0 and 1 to represent letters, numbers, and symbols used in computers and cell phones.	<table><tr><th>Number</th><th>Binary Code</th><th>Number</th><th>Binary Code</th></tr><tr><td>0</td><td>0</td><td>16</td><td>10000</td></tr><tr><td>1</td><td>01</td><td>17</td><td>10001</td></tr><tr><td>2</td><td>10</td><td>18</td><td>10010</td></tr><tr><td>3</td><td>11</td><td>19</td><td>10011</td></tr><tr><td>4</td><td>100</td><td>20</td><td>10100</td></tr><tr><td>5</td><td>101</td><td>21</td><td>10101</td></tr><tr><td>6</td><td>110</td><td>22</td><td>10110</td></tr><tr><td>7</td><td>111</td><td>23</td><td>10111</td></tr><tr><td>8</td><td>1000</td><td>24</td><td>11000</td></tr><tr><td>9</td><td>1001</td><td>25</td><td>11001</td></tr><tr><td>10</td><td>1010</td><td>26</td><td>11010</td></tr><tr><td>11</td><td>1011</td><td>27</td><td>11011</td></tr><tr><td>12</td><td>1100</td><td>28</td><td>11100</td></tr><tr><td>13</td><td>1101</td><td>29</td><td>11101</td></tr><tr><td>14</td><td>1110</td><td>30</td><td>11110</td></tr><tr><td>15</td><td>1111</td><td>31</td><td>11111</td></tr></table>	Number	Binary Code	Number	Binary Code	0	0	16	10000	1	01	17	10001	2	10	18	10010	3	11	19	10011	4	100	20	10100	5	101	21	10101	6	110	22	10110	7	111	23	10111	8	1000	24	11000	9	1001	25	11001	10	1010	26	11010	11	1011	27	11011	12	1100	28	11100	13	1101	29	11101	14	1110	30	11110	15	1111	31	11111
Number	Binary Code	Number	Binary Code																																																																			
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14	1110	30	11110																																																																			
15	1111	31	11111																																																																			

14

Identify **two** communication methods that use a pattern in order to transfer information. Use evidence from the table to support each choice. [1]

Method 1: _____

Evidence: _____

Method 2: _____

Evidence: _____

- 15 Using the information in the table, which statement about energy using the string-and-can telephone is most accurate?
- A Sound energy is transferred through the string.
 - B Electrical energy is transferred through the string.
 - C Sound energy is converted into electrical energy in the string.
 - D Heat energy is converted into sound energy in the can.
- 16 When drum signals are used for communication, what happens to the energy of the mallet, drum, and surrounding air when the mallet hits the drum?
- A The mallet gains energy from the drum and the surrounding air.
 - B The mallet and drum both gain energy from the surrounding air.
 - C The mallet loses energy, and this energy is transferred to the drum, but not to the surrounding air.
 - D The mallet loses energy, and this energy is transferred to the drum and then to the surrounding air.

Two friends live in buildings across the street from each other. They want to build their own method of communication that would allow them to find out if the other is available to meet outside. They can see and hear each other from their bedrooms. In order to be effective, this method must meet the following criteria and constraints.

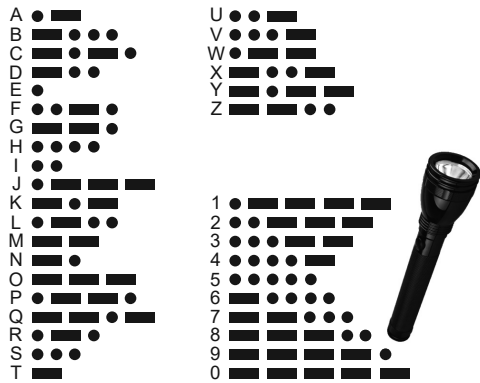
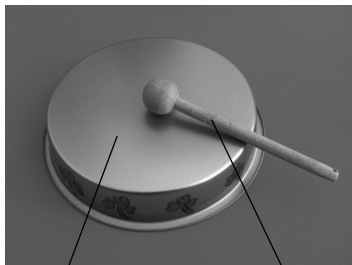
Criteria:

- be able to transfer information easily from one person to another
- inexpensive
- can be used either day or night

Constraints:

- must be quiet after 7:00 p.m.
- must be kept in their bedrooms
- must not take up a lot of space

They have proposed two different methods:

Method 1	Method 2
Morse code and flashlight 	Drum signals  Drum Mallet

17

Identify the method that allows the friends to *most* effectively receive each other's messages, and explain how this method best meets the criteria *and* constraints. [1]

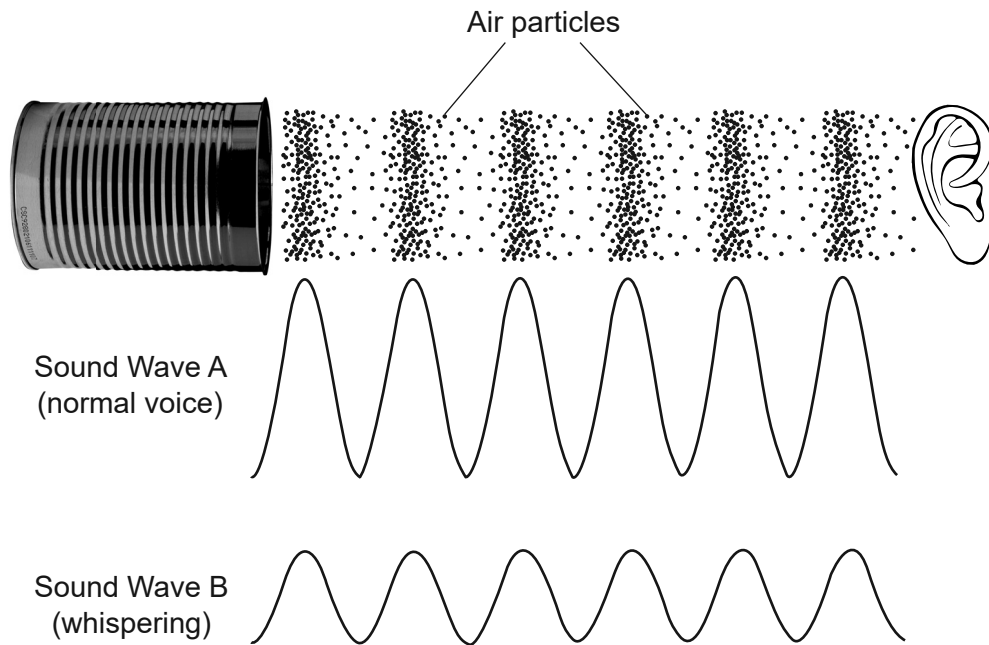
Method: _____

Explanation: _____

Air particles vibrate as a sound wave passes through them. The model below represents the motion of air particles caused by sound waves leaving a can and being heard by a person. The patterns of the particles can be changed into a model shaped like a wave.

Wave *A* represents a student speaking with their normal voice volume. Wave *B* represents the same student whispering. In both examples, the student spoke using the same pitch.

String-and-Can Telephone Waves



18

Compare the amplitude and wavelength of Wave *A* to the amplitude and wavelength of Wave *B*. Mention both waves in your responses. [1]

Amplitude: _____

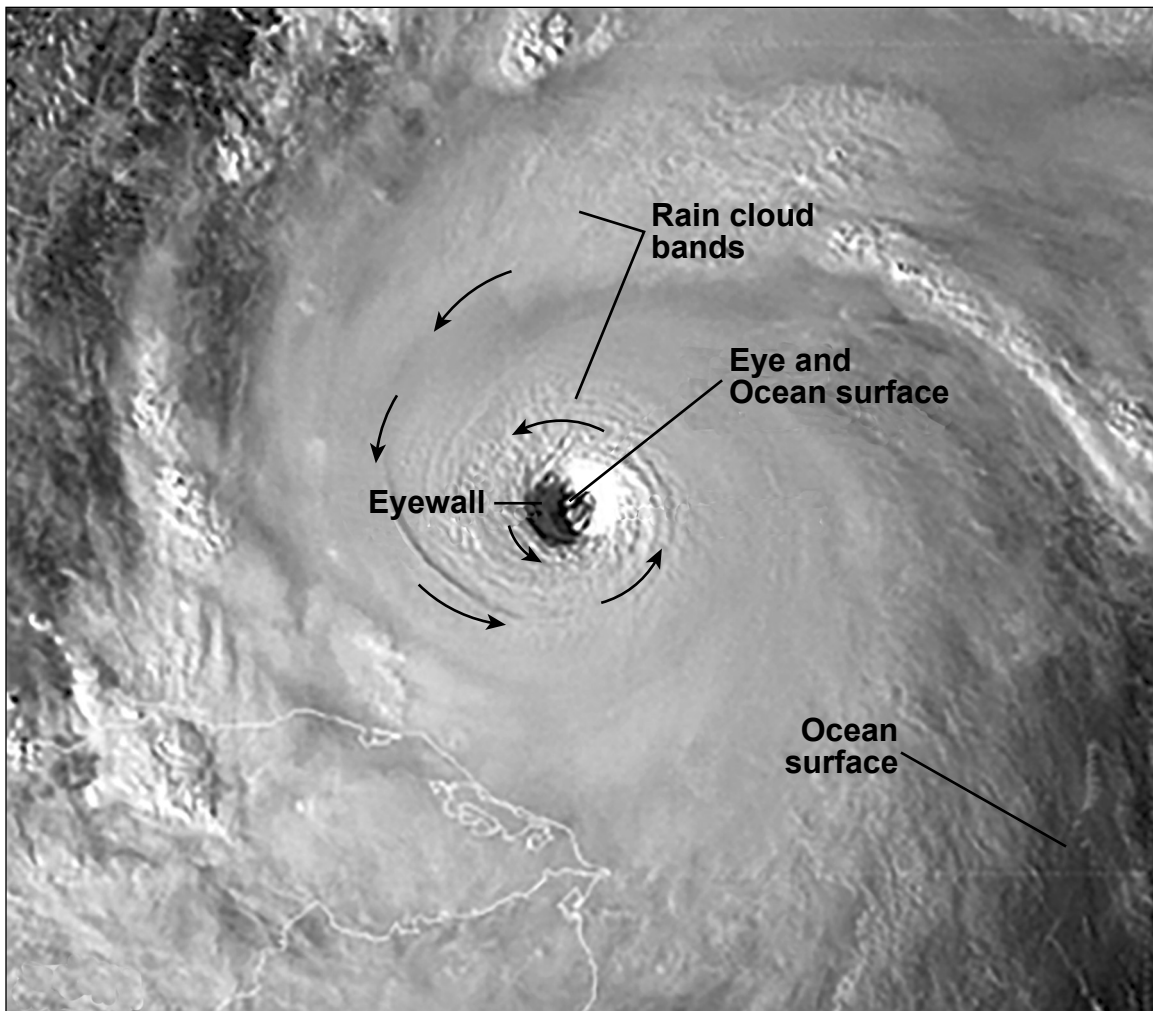
Wavelength: _____

Base your answers to questions 24 through 27 on the information below and on your knowledge of science.

Hurricane Formation

Hurricanes form from smaller storms. Heat energy in warm ocean water causes water vapor to enter the atmosphere. This vapor rises and cools, creating a storm system that forms clouds. Water falls back to Earth as rain. The storm starts to spin due to Earth's rotation. For storms to strengthen, there needs to be a continuous supply of warm ocean water. The storm spins around a center, called the eye. The winds in a hurricane are 74 miles per hour (mph) or greater. The strongest winds and thunderstorms occur at the edge of the eye, called the eyewall. Bands of rain clouds circle around the eye.

The photograph below shows a satellite image of a hurricane. Some parts of a hurricane are labeled. Arrows show the pattern of winds around the eye.



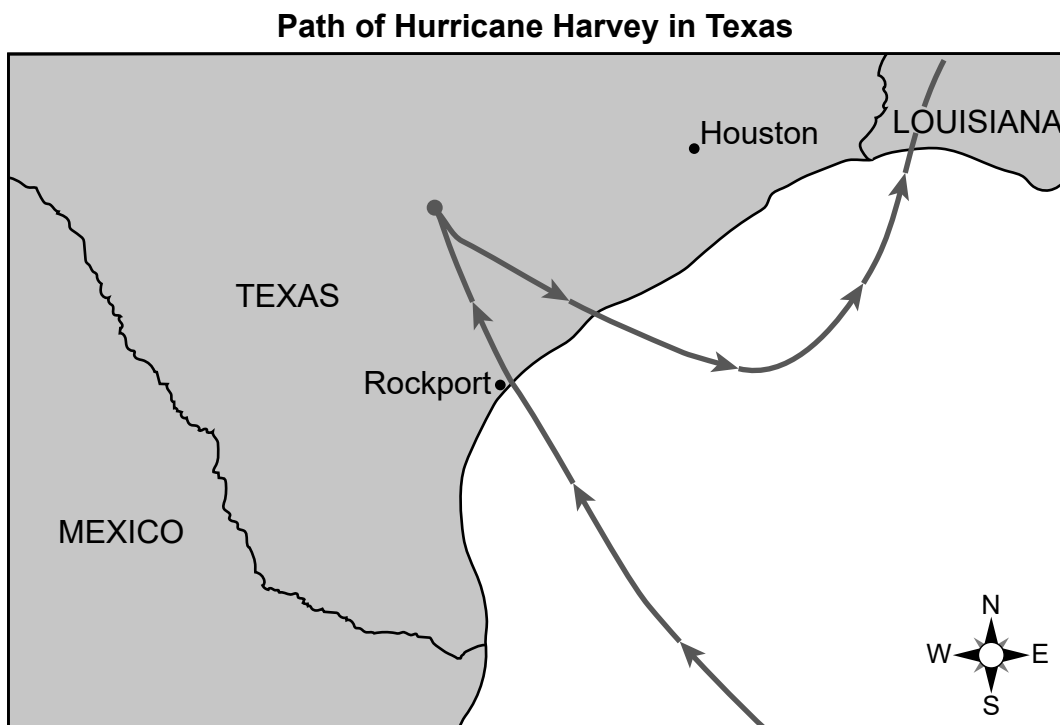
The table below describes water cycle processes that are the cause of or the effect of hurricane formation. Two processes are labeled 1 and 2.

Process	Description in Hurricane Formation
1	Heat energy in warm ocean water causes water vapor to enter the atmosphere.
2	Water vapor cools as it rises in the air, forming clouds.
precipitation	Rain falls back to Earth in bands of storms around the eye of the hurricane.
runoff	Water on Earth's surface from thunderstorms in hurricanes moves into lakes and rivers.

Which set of water cycle processes correctly completes the table?

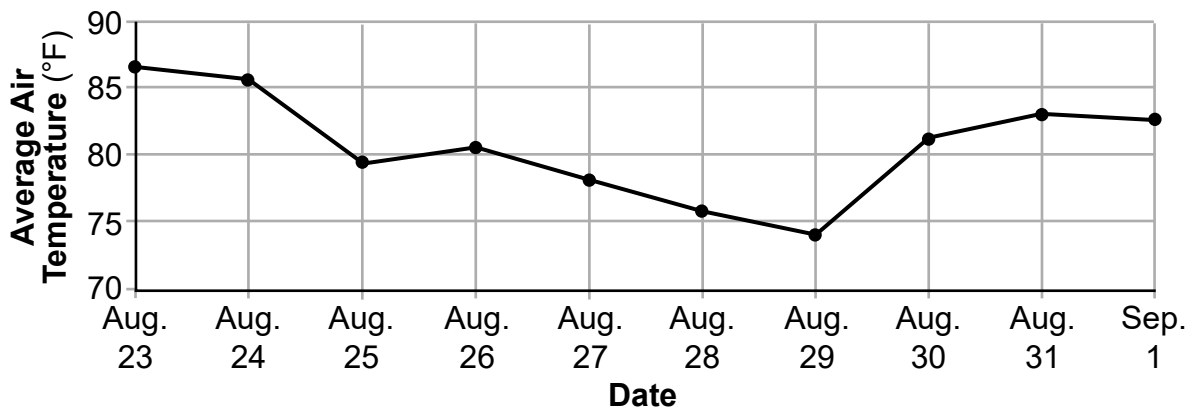
- A 1-groundwater flow 2-condensation
- B 1-evaporation 2-groundwater flow
- C 1-condensation 2-evaporation
- D 1-evaporation 2-condensation

Hurricane Harvey was a strong hurricane that made landfall on August 25, 2017 near Rockport, Texas. The map below shows the path of the hurricane as it moved toward central Texas and then moved east over a body of water. Arrows show the path of the hurricane. Harvey remained within Texas borders for 60 hours.

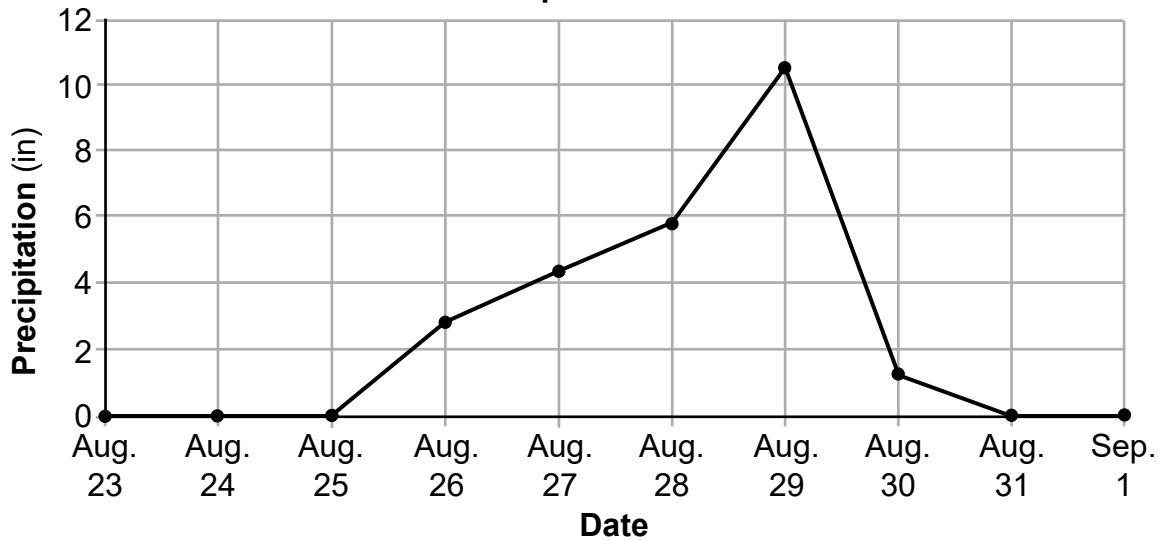


The graphs show some information about weather conditions in Houston, Texas in the last week in August, 2017.

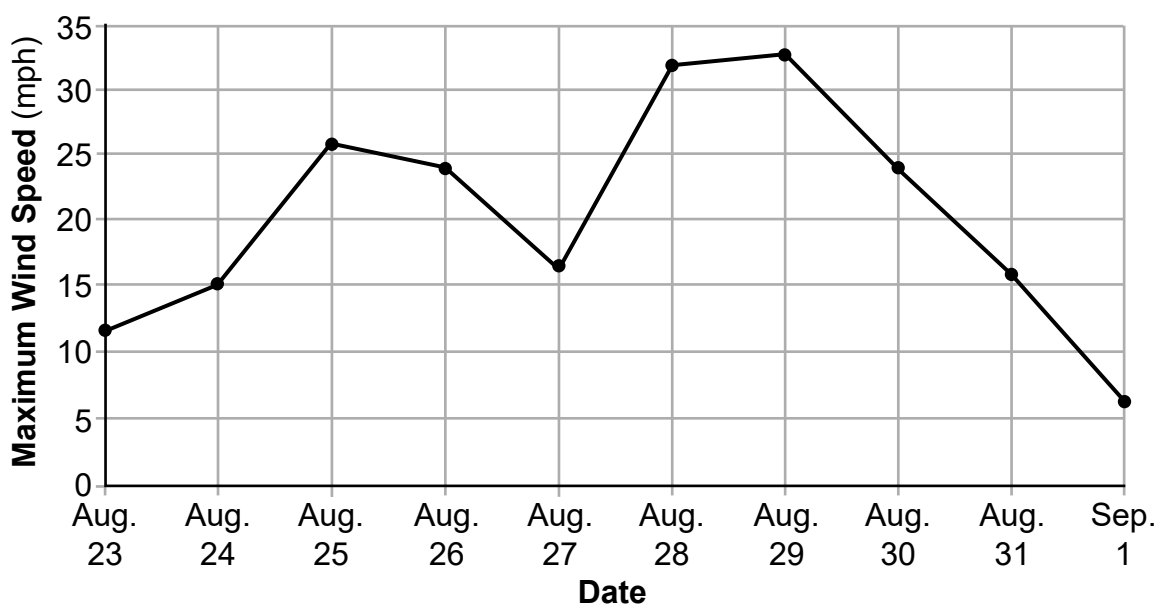
Average Air Temperature in Houston, Texas



Amount of Precipitation in Houston, Texas



Maximum Wind Speed in Houston, Texas



Based on the information in the graphs, select **one** date from the table during which the greatest effects of Hurricane Harvey in Houston, Texas were recorded.

Date

August 27, 2017	
August 28, 2017	
August 29, 2017	

Then, complete the passage below by placing the correct term in *each* box from the choices given. Terms may be used more than once. [1]

Term Choices:

decrease

increase

remain constant

Passage

Based on the patterns seen in the graphs, Hurricane Harvey caused the air

temperature to

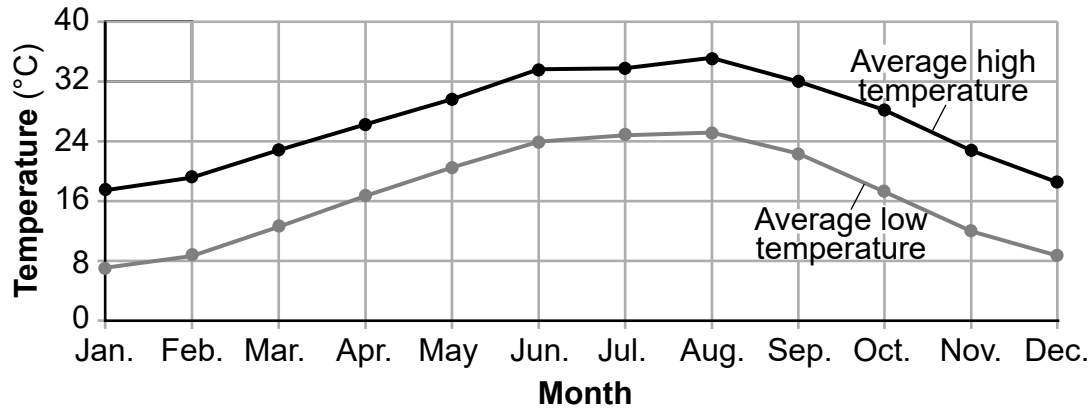
, the precipitation to

and the wind speed to

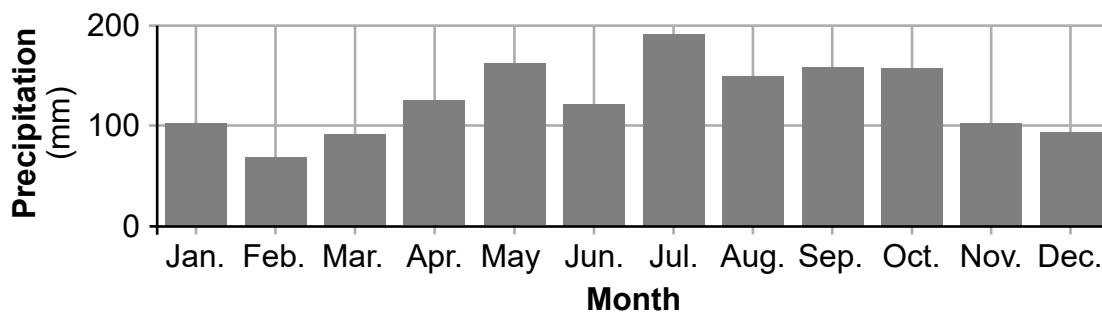
Hurricanes usually develop in the southern regions of the Atlantic Ocean instead of the northern regions of the Atlantic Ocean.

The graphs and table below show some information about the climate of Houston, Texas.

Monthly Average High and Low Air Temperatures in Houston, Texas



Monthly Average Precipitation in Houston, Texas



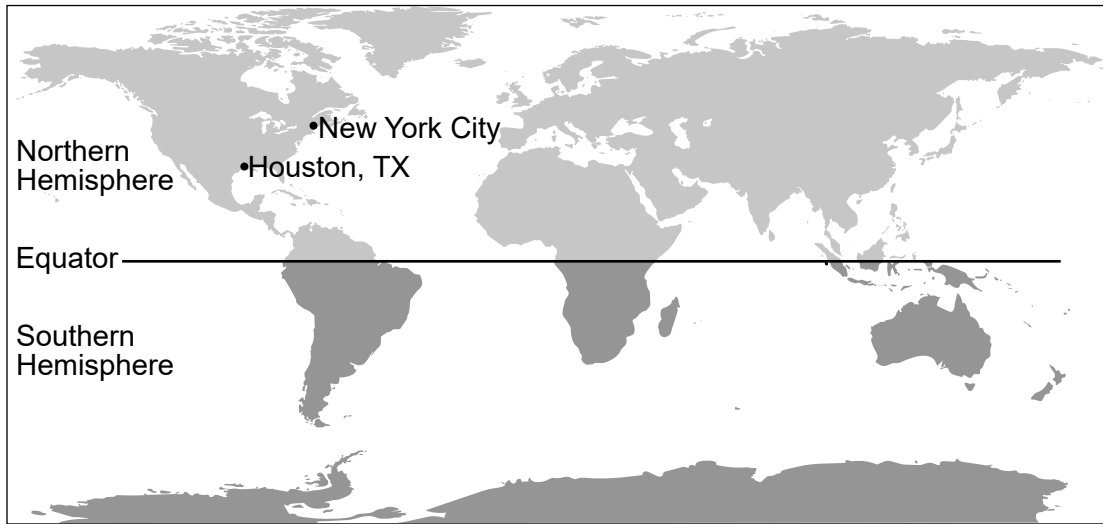
Monthly Average Surface Water Temperature of the Body of Water near Houston, Texas

Month	Jan.	Feb.	Mar.	Apr.	May	June	July	Aug.	Sept.	Oct.	Nov.	Dec.
Temperature (°F)	55.9	57.6	65.0	71.3	77.7	83.5	85.9	86.4	83.3	76.0	65.8	58.8

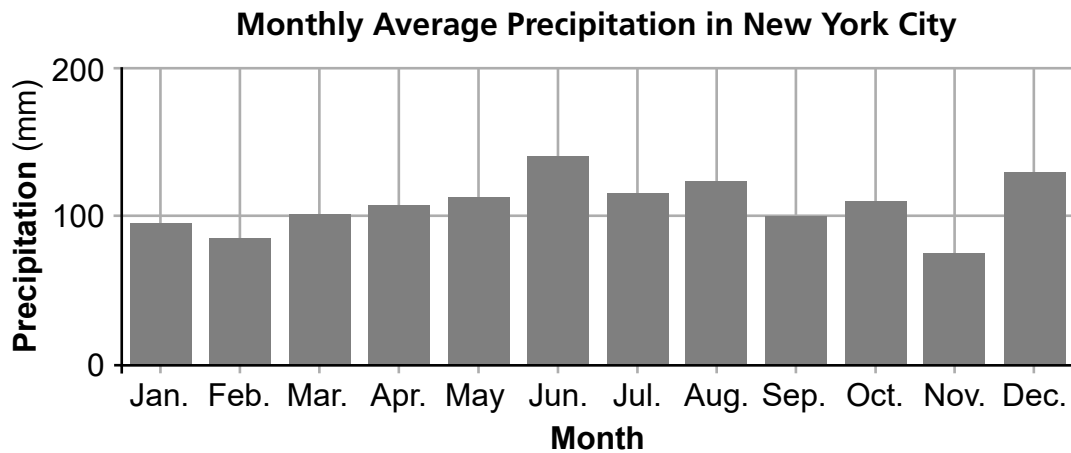
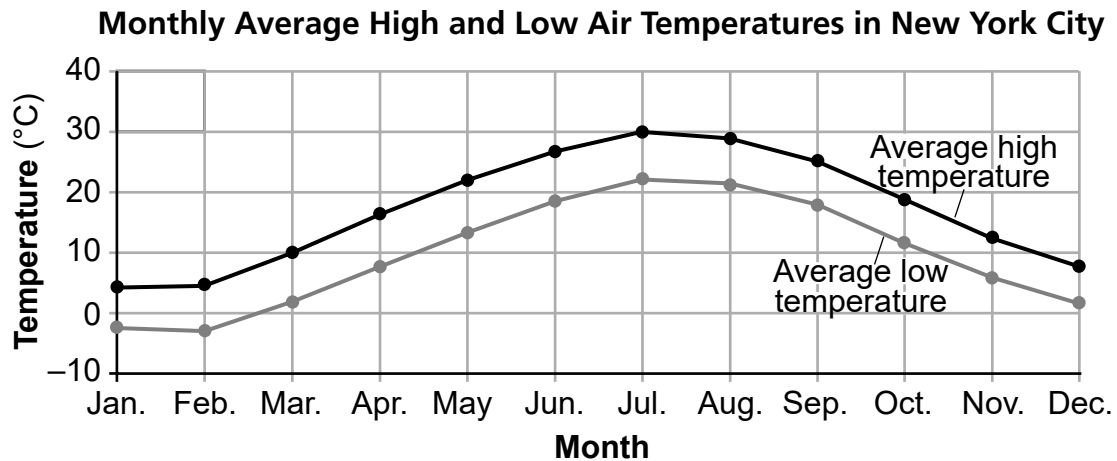
- 26** Hurricanes strengthen over the body of water near Houston, Texas. Based on the information provided, which statement correctly describes the reason this happens?
- A** In late summer months, the increase in precipitation near Houston supplies energy to further develop storms into hurricanes.
 - B** In late summer months, the surface water temperature increases enough to supply energy for the development of hurricanes.
 - C** In the spring, the monthly surface water temperature is at its highest, supplying energy to help develop storms into hurricanes.
 - D** In the spring, the monthly precipitation near Houston is at its highest, supplying energy to help develop storms into hurricanes.

The map below shows the location of Houston, Texas and New York City.

World Map



The graphs and table below show some information about the climate of New York City.



Monthly Average Surface Water Temperature of Atlantic Ocean near New York City

Month	Jan.	Feb.	Mar.	Apr.	May	June	July	Aug.	Sept.	Oct.	Nov.	Dec.
Temperature (°F)	40.6	38.4	41.0	47.7	56.9	66.1	72.1	74.9	72.4	64.4	54.4	46.2

- 27 Based on the information in the map and graphs, which statement correctly compares the locations and climate conditions of Houston, Texas, to New York City?
- A Houston is closer to the equator, has greater air temperatures in winter months, and more precipitation in December.
 - B Houston is closer to the equator, has greater air temperatures in summer months, and more precipitation in July.
 - C New York City is closer to the equator, has lower air temperatures in winter months, and less precipitation in July.
 - D New York City is closer to the equator, has lower air temperatures in summer months, and greater precipitation in December.

Base your answers to questions 28 through 31 on the information below and on your knowledge of science.

Locust Trees

Locust trees are native to North America. They are found in various locations across the United States. There are many different types of locust trees, such as the honey locust. This tree gets its name from the honey-like substance found in its seed pods. The sweet seed pods of honey locusts are an important food source. White-tailed deer, rabbits, squirrels, and foxes will eat the honey locust seed pods. Deer will also eat the bark of the honey locust tree in the winter when other food sources are scarce.

Honey Locust Tree



Seed Pod of a Honey Locust Tree



28

Explain how the taste of the honey locust seed pods contributes to the reproduction of the honey locust trees. [1]

Many mature honey locust trees have long thorns on their trunks and smaller thorns on their branches. Not all honey locust trees have thorns. The thornless variety of this tree has similar seed pods, leaves, and flowers to the variety with thorns. Honey locust trees are often planted in cities due to their ability to survive in poor air quality and periods of dryness. Seeds from honey locust trees with thorns may grow into trees without thorns. Likewise, seeds from honey locust trees without thorns may grow into trees with thorns.

**Trunk of Honey Locust
Tree without Thorns**



**Trunk of Honey Locust
Tree with Thorns**



29 Which statement describes a variation in a trait that exists in honey locust trees?

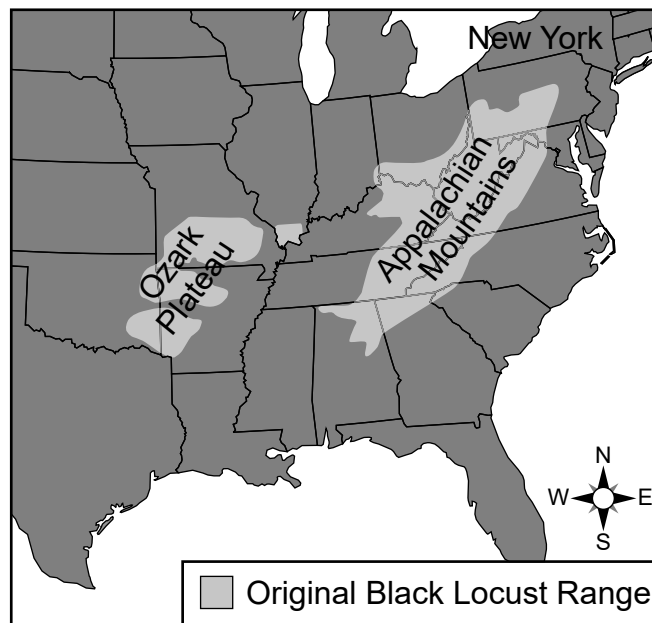
- A Honey locust trees are often planted in cities.
- B All honey locust trees have seed pods.
- C Some honey locust trees have thorns.
- D Honey locust trees can live in areas with poor air quality.

30 Which argument supports the claim that honey locust trees with thorns are more likely to survive than honey locust trees without thorns?

- A The thorns of these trees will produce more food for the tree.
- B The thorns of these trees help absorb more water for growth.
- C The thorns of these trees provide habitats for moths and caterpillars.
- D The thorns of these trees make it difficult to eat the tree's bark.

The black locust is a tree similar to the honey locust. The map below shows the original range for the black locust in the United States. This range was primarily in the Appalachian Mountains and the Ozark Plateau.

Original Black Locust Range



The distribution of the black locust has now spread in the United States to the Midwest and Northeast, including New York State. This tree has a fast growth rate and grows in groups. These groups decrease land surface temperatures. The groups also decrease the amount of light that can reach low-growing plants of grasslands and prairies.

31

Which set of terms correctly completes the passage below?

The introduction of the black locust to the Midwest prairie ecosystem could ____ 1 ____ the amount of grass and clover, resulting in a(n) ____ 2 ____ in the number of grasshoppers and owls because the black locust could ____ 3 ____ the balance of the ecosystem.

A

1	increase
2	decrease
3	upset

C

1	decrease
2	increase
3	improve

B

1	decrease
2	decrease
3	upset

D

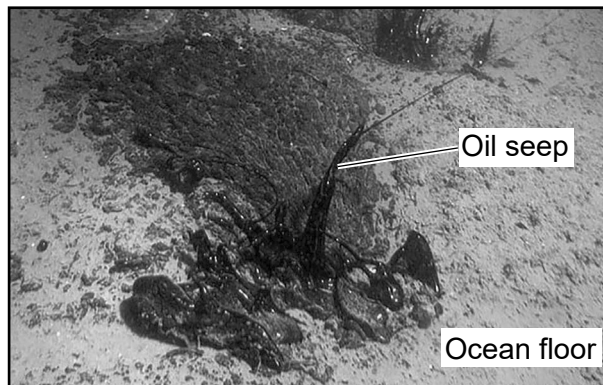
1	increase
2	increase
3	improve

Base your answers to questions 32 through 35 on the information below and on your knowledge of science.

Natural Energy Resources

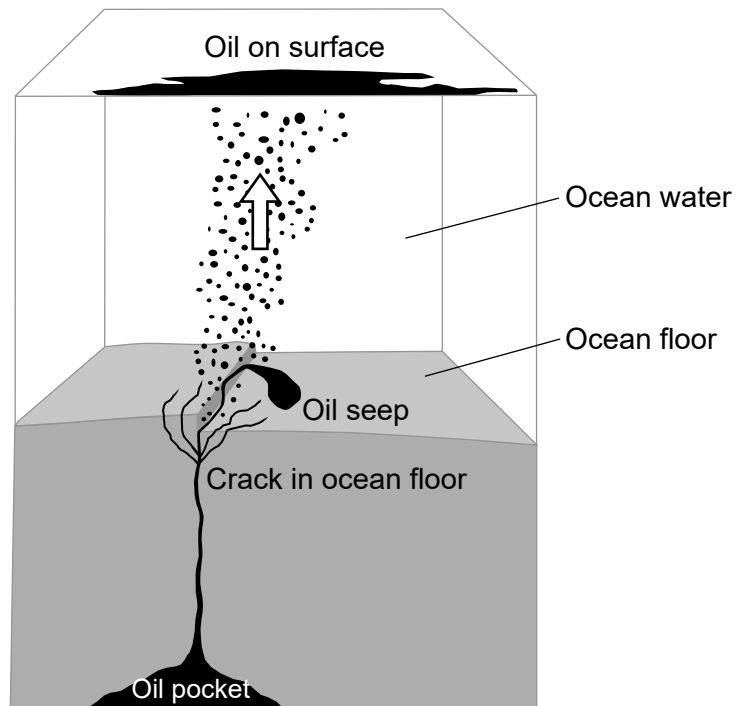
Humans use many types of natural resources to provide energy for their daily lives. Petroleum, also called crude oil or oil, is a fossil fuel formed millions of years ago. It is an important fuel source used around the world. Oil can be refined to produce heating fuel and gasoline. It can also be used to produce electrical energy.

Oil is found in pockets under Earth's surface. The photograph below shows an oil seep. Oil seeps occur when crude oil leaks through natural cracks in the ocean floor.



The model below represents the positions of both the oil pocket and oil seep in relation to the ocean floor. The direction of motion of oil droplets in the ocean is indicated.

Model of Movement of Oil in an Oil Seep



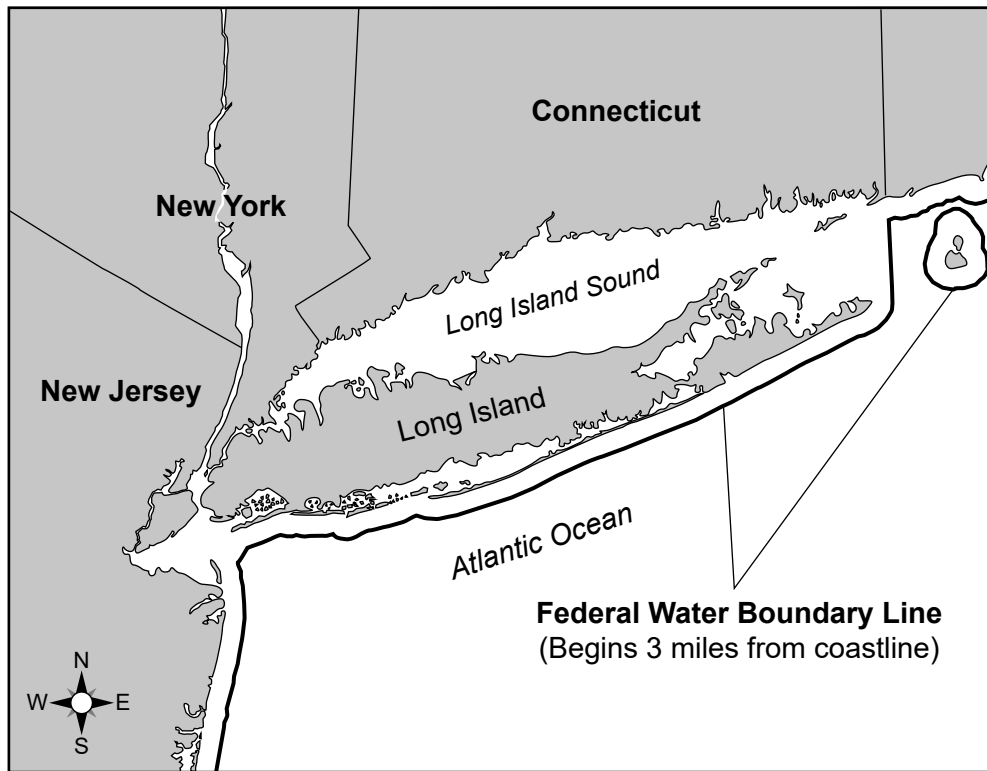
32

Which two Earth spheres are shown interacting in the model as a result of this oil seep?

- A hydrosphere and atmosphere
- B biosphere and geosphere
- C geosphere and hydrosphere
- D atmosphere and biosphere

Offshore drilling is a way of obtaining underground oil from Earth by drilling into the ocean floor where oil pockets have been found. In 2019, New York State passed a law banning offshore drilling in the ocean floor within three miles of the state's shoreline.

New York's Marine and Coastal District Waters



33

Offshore drilling can have negative effects on the environment. Describe **one negative** effect that may have led New York State to ban offshore drilling. [1]

Wind is another natural resource that humans use to provide energy for their daily lives. Wind is used to produce electricity with large windmills called wind turbines. A wind farm is a group of wind turbines in the same location. The photograph below shows Maple Ridge Wind Farm, located in Lewis County, New York.

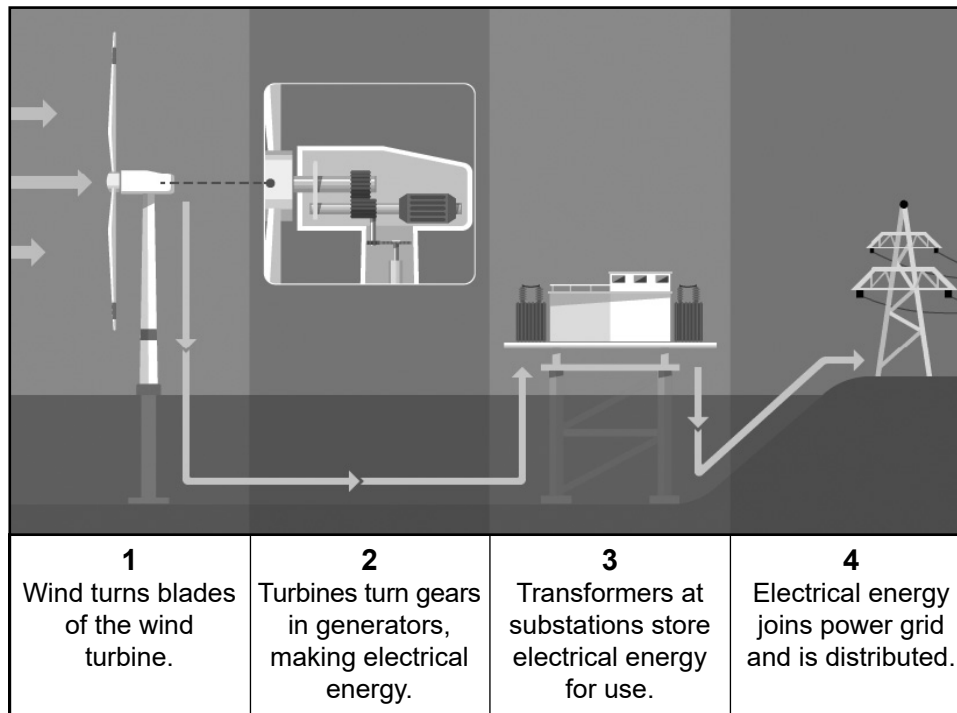


34

Describe how the Maple Ridge Wind Farm helps protect the environment. [1]

The model below shows how a wind turbine functions to produce electrical energy.

Model of How a Wind Turbine Works



35 What type of energy is the wind turbine converting into electrical energy?

- A light
- B heat
- C motion
- D sound

THE STATE EDUCATION DEPARTMENT
THE UNIVERSITY OF THE STATE OF NEW YORK / ALBANY, NY 12234
 2026 Elementary-level Science Test Map to the Standards
 Grade 5 Released Questions

Question	Type	Key	Points	Performance Expectation	Subscore	Percentage of Students Who Answered Correctly (P-Value)
1	Constructed Response		1	4-PS4-2	PS	
2	Multiple Choice	D	1	5-ESS1-2	ESS	
3	Multiple Choice	A	1	5-ESS1-2	ESS	
4	Constructed Response		1	5-ESS1-2	ESS	
5	Multiple Choice	A	1	5-PS1-3	PS	
6	Constructed Response		1	5-PS1-3	PS	
7	Multiple Choice	B	1	5-PS1-2	PS	
8	Constructed Response		1	5-PS1-4	PS	
9	Multiple Choice	A	1	5-PS1-3	PS	
10	Multiple Choice	A	1	3-LS2-1	LS	
11	Multiple Choice	B	1	3-LS4-2	LS	
12	Constructed Response		1	5-LS2-1	LS	
13	Multiple Choice	C	1	3-LS4-3	LS	
14	Constructed Response		1	4-PS4-3	PS	
15	Multiple Choice	A	1	4-PS3-2	PS	
16	Multiple Choice	D	1	4-PS3-3	PS	
17	Constructed Response		1	3-5-ETS1-2		
18	Constructed Response		1	4-PS4-1	PS	
24	Multiple Choice	D	1	3-ESS2-3	ESS	
25	Constructed Response		1	3-ESS2-1	ESS	
26	Multiple Choice	B	1	3-ESS2-1	ESS	
27	Multiple Choice	B	1	3-ESS2-2	ESS	
28	Constructed Response		1	4-LS1-1	LS	
29	Multiple Choice	C	1	3-LS3-1	LS	
30	Multiple Choice	D	1	4-LS1-1	LS	
31	Multiple Choice	B	1	5-LS2-1	LS	
32	Multiple Choice	C	1	5-ESS2-1	ESS	
33	Constructed Response		1	4-ESS3-1	ESS	
34	Constructed Response		1	5-ESS3-1	ESS	
35	Multiple Choice	C	1	4-PS3-2	PS	

* This item map identifies the Performance Expectation with which each test question is aligned. All NYSP-12SLS Performance Expectations are three-dimensional (<https://www.nysed.gov/sites/default/files/programs/standards-instruction/p-12-science-learning-standards.pdf>). The integration of these three dimensions provides students with a context for the content of science (DCI), the methods by which science knowledge is acquired and understood (SEP), and the ways in which the sciences are connected through concepts that have universal meaning across the disciplines (CCC).

It is recommended that prior to scoring student responses to a specific question, raters should review the entire cluster, including all associated stimuli (e.g., reading passages, data tables, graphs, and diagrams).

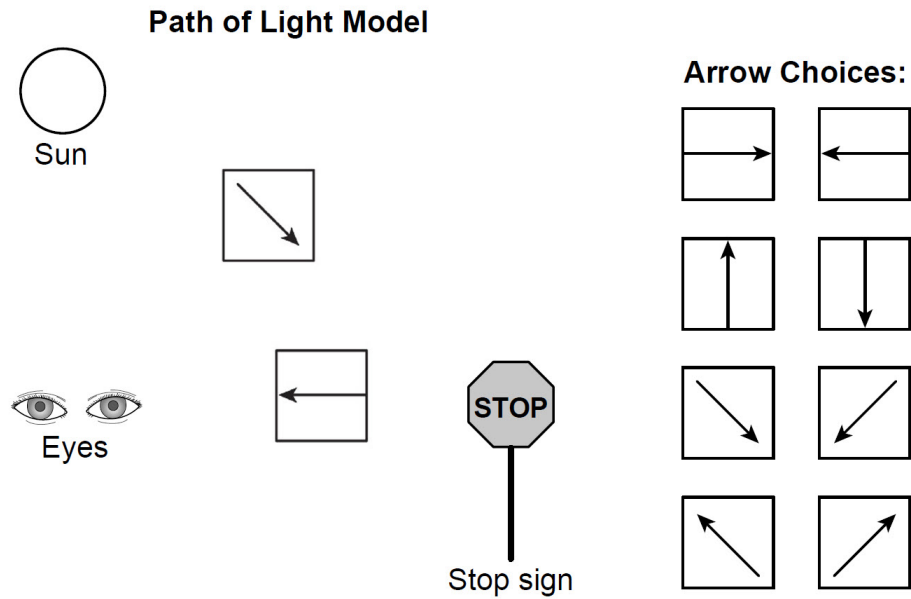
Constructed-Response Science Scoring Policies

1. Examples of acceptable responses are provided; however, other scientifically accurate responses that demonstrate adequate knowledge may receive credit. Raters must use their professional judgment to decide if the student's response meets the criteria.
2. If the student provides a response that contains errors in spelling, grammar, capitalization, or punctuation, the student should still receive credit, provided the errors do not affect the scientific accuracy of the response.
3. If the student provides a response that contains a correct answer along with extraneous information, the student should still receive credit, provided the extraneous information does not negate or contradict the correct response.
4. For questions with multiple components (e.g., a technology enhanced portion and a constructed-response portion), students must answer all components correctly to receive credit.
5. Condition Code A is applied whenever a student who is present for a test session leaves an entire constructed-response question completely blank (no response attempted). This is not to be confused with a score of zero wherein the student does respond to part or all of the question, but that work results in a score of zero.

EXEMPLARY RESPONSE

1

Sunlight allows objects to be visible on Earth. From the choices, place one arrow in *each* box to complete the model to show the correct path of light that permits a stop sign to be seen. [1]



EXEMPLARY RESPONSE

4

From the choices below, place **one** correct constellation name in *each* box to indicate the constellations that would be entirely visible and would ***not*** be visible from Rochester, New York at night on March 21. [1]

Visible

Libra

OR

Virgo

**Choices for
Constellations:**

Gemini

Libra

Pisces

Sagittarius

Virgo

Not Visible

Pisces

OR

Sagittarius

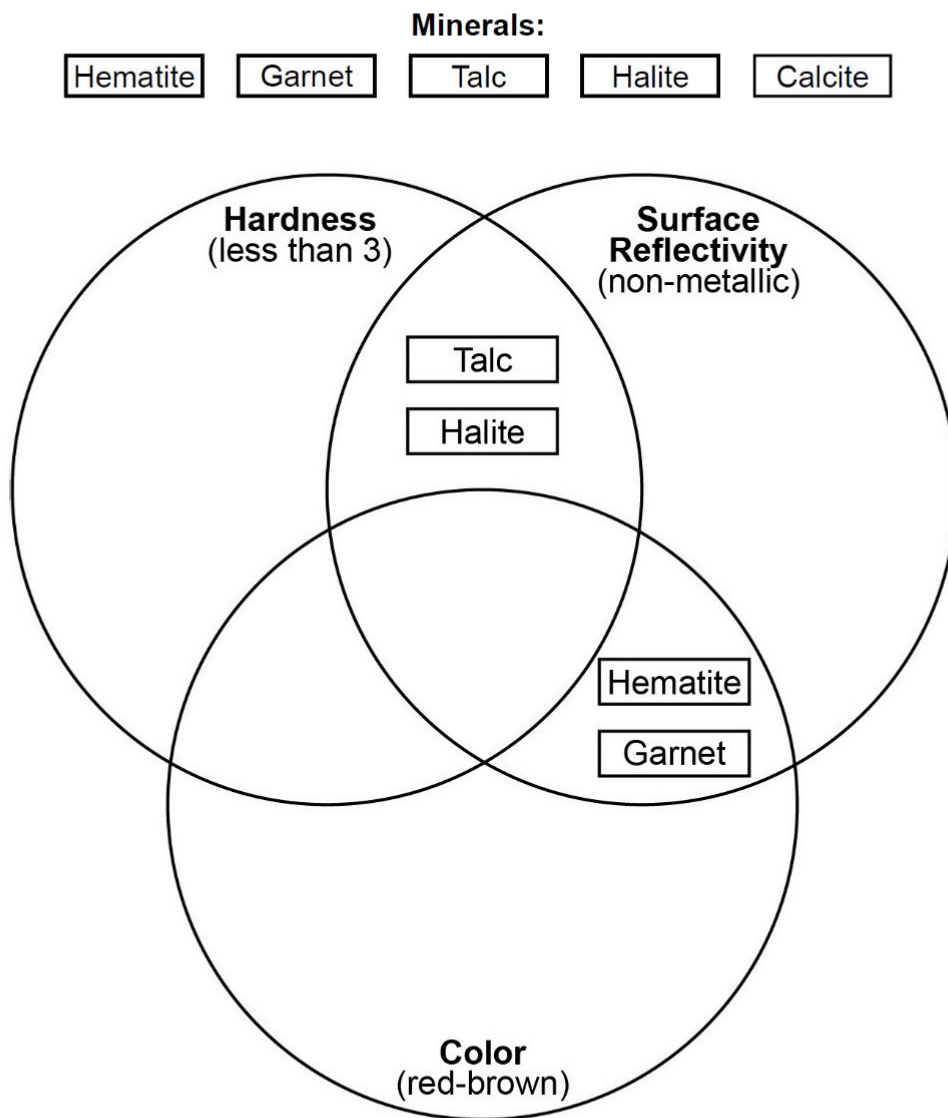
OR

Gemini

EXEMPLARY RESPONSE

6

The model represents how three properties can be used to classify these minerals. The areas of overlap show shared properties. Based on the properties shown in *Minerals of New York State and Their Observed Properties*, place the names of **four** of the minerals from the choices below in the correct boxes within the model. [1]



OR a complete model with the same mineral names indicated in a different order within the same region of the model

EXEMPLARY RESPONSE

8

A student claimed that one of the beakers showed evidence that the mixing of substances resulted in a new substance. Identify the beaker number in which a new substance was formed. Then identify the evidence in the model that supports your choice. [1]

Beaker number: 2/two/the second one

Evidence in model:

The calcite pieces bubbled in vinegar, which means there is a new substance.

OR

Bubbles of gas were produced from the calcite pieces in vinegar. The gas is the new substance.

OR

Bubbles/gas formed

OR other valid response

ELS Training 1

8

A student claimed that one of the beakers showed evidence that the mixing of substances resulted in a new substance. Identify the beaker number in which a new substance was formed. Then identify the evidence in the model that supports your choice. [1]

Beaker number: _____ 2 _____

Evidence in model:

Beaker number two has created a new substance because there turned out to be a gas forming in bubbles.

Score Credit 1 (out of 1 credit)

This response demonstrates a clear understanding of the concepts in the task.

- Beaker 2 is chosen.
- The response correctly identifies the appearance of a gas as evidence of a new substance being formed.

This response is complete and correct.

ELS Training 2

8

A student claimed that one of the beakers showed evidence that the mixing of substances resulted in a new substance. Identify the beaker number in which a new substance was formed. Then identify the evidence in the model that supports your choice. [1]

Beaker number: _____ 2 _____

Evidence in model:

Number 2 is a new substance because of the bubbles.

Score Credit 1 (out of 1 credit)

This response demonstrates a fair understanding of the concepts in the task.

- Beaker 2 is chosen.
- The response correctly identifies the appearance of bubbles as evidence of a new substance being formed.

This response is sufficient for credit.

ELS Training 3

8

A student claimed that one of the beakers showed evidence that the mixing of substances resulted in a new substance. Identify the beaker number in which a new substance was formed. Then identify the evidence in the model that supports your choice. [1]

Beaker number: _____ 2 _____

Evidence in model:

It started forming bubbles.

Score Credit 1 (out of 1 credit)

This response demonstrates a fair understanding of the concepts in the task.

- Beaker 2 is chosen.
- The response correctly identifies the appearance of bubbles as evidence of a new substance being formed.

This response is sufficient for credit.

ELS Training 4

8

A student claimed that one of the beakers showed evidence that the mixing of substances resulted in a new substance. Identify the beaker number in which a new substance was formed. Then identify the evidence in the model that supports your choice. [1]

Beaker number: two

Evidence in model:

Beaker two is a new substance because air was made.

Score Credit 0 (out of 1 credit)

This response demonstrates incomplete understanding of the concepts in the task.

- Beaker 2 is chosen.
- The response incorrectly describes the bubbles as air rather than gas.

This response is insufficient for credit due to a lack of scientific accuracy.

ELS Training 5

8

A student claimed that one of the beakers showed evidence that the mixing of substances resulted in a new substance. Identify the beaker number in which a new substance was formed. Then identify the evidence in the model that supports your choice. [1]

Beaker number: _____ 2 _____

Evidence in model:

They are showing the Calcite pieces mixed in with some bubbles and liquid which creates a new substance.

Score Credit 0 (out of 1 credit)

This response demonstrates incomplete understanding of the concepts in the task.

- Beaker 2 is chosen.
- The response demonstrates a misunderstanding of the bubbles as a reactant and not as the new substance formed.

This response is insufficient for credit due to a lack of scientific accuracy.

ELS Training 6

8

A student claimed that one of the beakers showed evidence that the mixing of substances resulted in a new substance. Identify the beaker number in which a new substance was formed. Then identify the evidence in the model that supports your choice. [1]

Beaker number: _____ 2 _____

Evidence in model:

liquid bubbles calcite pieces vinegar 30g calcite 10g

Score Credit 0 (out of 1 credit)

This response demonstrates no overall understanding of the concepts in the task.

- Beaker 2 is chosen.
- The response lists everything in the beaker directly from the model and fails to identify bubbles as the new substance formed.

This response is insufficient for credit due to a lack of supporting evidence.

EXEMPLARY RESPONSE

12

Identify **one** food source for the white-tailed deer in Yellowstone National Park. Then, describe how this food source would be affected if the white-tailed deer population changed when more wolves were introduced into the park. Include a reason for this change in your description. [1]

Food source: **cottonwood/willow**

Effect on food source:

The cottonwood (or willow) would probably increase because more deer would be killed by the wolves, so the deer would not be eating the plants.

OR

There would be more willow (or cottonwoods) because less deer would be eating them due to more wolves eating more deer.

OR other valid response

ELS Training 1

12

Identify **one** food source for the white-tailed deer in Yellowstone National Park. Then, describe how this food source would be affected if the white-tailed deer population changed when more wolves were introduced into the park. Include a reason for this change in your description. [1]

Food source: _____ **Willow** _____

Effect on food source:

If more wolves are introduced to yellowstone, there will be less White-tailed deer to eat willow because of the increase of the deer's predator. There will then be more willow since less animals will be eating the plant.

Score Credit 1 (out of 1 credit)

This response demonstrates a clear understanding of the concepts in the task.

- Willow is chosen as the food source.
- A description that the food source would increase is provided.
- A reason for this increase is provided.

This response is complete and correct.

ELS Training 2

12

Identify **one** food source for the white-tailed deer in Yellowstone National Park. Then, describe how this food source would be affected if the white-tailed deer population changed when more wolves were introduced into the park. Include a reason for this change in your description. [1]

Food source: _____ cottonwood _____

Effect on food source:

The deer would be prey to wolves and so then there will be a decrease of deer. Cottonwood's population would increase. This would happen because less deer means more cottonwood.

Score Credit 1 (out of 1 credit)

This response demonstrates a fair understanding of the concepts in the task.

- Cottonwood is chosen as the food source.
- A description that the food source would increase is provided.
- A reason for this increase is minimally provided, as it must be inferred that the deer eat the cottonwood.

This response is sufficient for credit.

ELS Training 3

12

Identify **one** food source for the white-tailed deer in Yellowstone National Park. Then, describe how this food source would be affected if the white-tailed deer population changed when more wolves were introduced into the park. Include a reason for this change in your description. [1]

Food source: _____ **Willow** _____

Effect on food source:

If more wolves come to Yellowstone, then more willow will not be eaten because more white tailed deer are being hunted.

Score Credit 1 (out of 1 credit)

This response demonstrates a fair understanding of the concepts in the task.

- Willow is chosen as the food source.
- A description implying that the food source would increase is provided.
- A reason for this increase is provided.

This response is sufficient for credit.

ELS Training 4

12

Identify **one** food source for the white-tailed deer in Yellowstone National Park. Then, describe how this food source would be affected if the white-tailed deer population changed when more wolves were introduced into the park. Include a reason for this change in your description. [1]

Food source: Cotton Wood

Effect on food source:

I picked the Cotton Wood. The amount of Cotton Wood's that would change would not go down as fast. I know this because the wolf's would kill the deer so the Cotton Wood's would not be affected as much by the deer since the deer are dead.

Score Credit 0 (out of 1 credit)

This response demonstrates incomplete understanding of the concepts in the task.

- Cottonwood is chosen as the food source.
- The description that the cottonwood “would not go down as fast” is ambiguous and does not imply an increase in the food source due to a disruption to the ecosystem.
- A reason for the change is provided.

This response is insufficient for credit due to an incomplete description.

ELS Training 5

12

Identify **one** food source for the white-tailed deer in Yellowstone National Park. Then, describe how this food source would be affected if the white-tailed deer population changed when more wolves were introduced into the park. Include a reason for this change in your description. [1]

Food source: _____ cottonwood _____

Effect on food source:

If more wolves came into yellowstone national park they would eat the white tailed deer and then not many of the white tailed deers would be able to eat the cotton wood some maybe the food source for white tailed deers change.

Score Credit 0 (out of 1 credit)

This response demonstrates incomplete understanding of the concepts in the task.

- Cottonwood is chosen as the food source.
- The description that the food source would “change” does not imply an increase.
- A reason for the change is provided.

This response is insufficient for credit due to an incomplete description.

ELS Training 6

12

Identify **one** food source for the white-tailed deer in Yellowstone National Park. Then, describe how this food source would be affected if the white-tailed deer population changed when more wolves were introduced into the park. Include a reason for this change in your description. [1]

Food source: _____ **cottonwood** _____

Effect on food source:

There would be more cottonwood for elk and beavers to eat.

Score Credit 0 (out of 1 credit)

This response demonstrates incomplete understanding of the concepts in the task.

- Cottonwood is chosen as the food source.
- A description that the food source would increase is provided.
- No reason for this increase is provided.

This response is insufficient for credit due to an incomplete description.

EXEMPLARY RESPONSE

14

Identify **two** communication methods that use a pattern in order to transfer information. Use evidence from the table to support each choice. [1]

Method 1: **Drum Signals**_____

Evidence: **Sound patterns made by a drum communicate information.**

Method 2: **Morse Code**_____

Evidence: **Patterns of long and short flashing lights or dots and dashes to symbolize letters and numbers.**

OR

Identify **two** communication methods that use a pattern in order to transfer information. Use evidence from the table to support each choice. [1]

Method 1: **Drum Signals**_____

Evidence: **Information is transferred by patterns of drum beats.**

Method 2: **Sign Language**_____

Evidence: **Making the same hand movement represents the same letter/word every time.**

OR

Identify **two** communication methods that use a pattern in order to transfer information. Use evidence from the table to support each choice. [1]

Method 1: **Morse Code**_____

Evidence: **Is a pattern of sounds that transfers information.**

Method 2: **Binary Code**_____

Evidence: **Patterns made by 0s and 1s represent letters and numbers.**

OR other valid response

ELS Training 1

14

Identify **two** communication methods that use a pattern in order to transfer information. Use evidence from the table to support each choice. [1]

Method 1:

Evidence:

there is a pattern in the drum signals because they can play different lengths of sounds or tones and there can be different pitches in the sounds or in the tones

Method 2:

Evidence:

With morse there is a pattern of short sounds or lights or dots and long sounds lights or dots to represent different letters and numbers to show what the communicator is trying to get across

Score Credit 1 (out of 1 credit)

This response demonstrates a clear understanding of the concepts in the task.

- Drum signals are chosen as the first method and the evidence provided describes its patterns.
- Morse code is chosen as the second method and the evidence provided explains how its patterns are used to communicate.

This response is complete and correct.

ELS Training 2

14

Identify **two** communication methods that use a pattern in order to transfer information. Use evidence from the table to support each choice. [1]

Method 1: Binary code

Evidence: It is a pattern of 1s and 0s

Method 2: Morse code

Evidence: It is a pattern of dots and dashes

Score Credit 1 (out of 1 credit)

This response demonstrates a fair understanding of the concepts in the task.

- Binary code is chosen as the first method and the evidence provided minimally describes its pattern.
- Morse code is chosen as the second method and the evidence provided minimally describes its pattern.

This response is sufficient for credit.

ELS Training 3

14

Identify **two** communication methods that use a pattern in order to transfer information. Use evidence from the table to support each choice. [1]

Method 1: Morse code

Evidence:

It uses dot and sounds which has a repeating pattern we can see on the picture that is provided to us.

Method 2: Binary code

Evidence:

It uses 1 and 0 to represent a code and send a message.

Score Credit 1 (out of 1 credit)

This response demonstrates a fair understanding of the concepts in the task.

- Morse code is chosen as the first method and the evidence provided minimally describes its pattern.
- Binary code is chosen as the second method and the evidence provided minimally describes its pattern.

This response is sufficient for credit.

ELS Training 4

14

Identify **two** communication methods that use a pattern in order to transfer information. Use evidence from the table to support each choice. [1]

Method 1: Binary code

Evidence:

the numbers are the same but it switches up every time

Method 2: drum signals

Evidence:

it is a system that uses different sound combinations so it has to be a pattern

Score Credit 0 (out of 1 credit)

This response demonstrates incomplete understanding of the concepts in the task.

- Binary code is chosen as the first method, but the evidence provided does not describe its pattern.
- Drum signals is chosen as the second method and the evidence provided minimally describes its pattern.

This response is insufficient for credit due to lack of support for Method 1.

ELS Training 5

14

Identify **two** communication methods that use a pattern in order to transfer information. Use evidence from the table to support each choice. [1]

Method 1: Morse code

Evidence: "System that uses long and short sounds, long or short flashing lights, or dots and dashes, to represent letters and number."

Method 2: Binary code

Evidence: "System that uses 0 and 1 to represent letter, numbers, and symbols used in computers and cell phones."

Score Credit 0 (out of 1 credit)

This response demonstrates incomplete understanding of the concepts in the task.

- Morse code is chosen as the first method, but the evidence provided is a direct copy from the table and does not describe its pattern.
- Binary code is chosen as the second method, but the evidence provided is a direct copy from the table and does not describe its pattern.

This response is insufficient for credit due to lack of support.

ELS Training 6

14

Identify **two** communication methods that use a pattern in order to transfer information. Use evidence from the table to support each choice. [1]

Method 1: string and can telephone

Evidence: one person talks and then another prson talks it keeps on going in a pattern

Method 2: morse code

Evidence: people use a flash light and a person and the other person does it it goes in a pattern

Score Credit 0 (out of 1 credit)

This response demonstrates incomplete understanding of the concepts in the task.

- String-and-can telephone is chosen as the first method, but the evidence provided describes the back-and-forth pattern of conversation, not a pattern to transfer information.
- Morse code is chosen as the second method, but the evidence provided describes the back-and-forth pattern of conversation, not a pattern to transfer information.

This response is insufficient for credit due to lack of support.

ELS Training 7

14

Identify **two** communication methods that use a pattern in order to transfer information. Use evidence from the table to support each choice. [1]

Method 1: Ticking

Evidence:

You can make different tick noises in your mouth to communicate.

Method 2: Braille

Evidence:

You use this if you are blind. You put your fingers across dots and the dots equal a letter that equals a word.

Score Credit 0 (out of 1 credit)

This response demonstrates incomplete understanding of the concepts in the task.

- Ticking is chosen as the first method, but the evidence provided does not describe a pattern and does not come from the table.
- Braille is chosen as the second method, but the evidence provided does not come from the table.

This response is insufficient for credit because the methods and supporting evidence do not come from the table.

EXEMPLARY RESPONSE

17

Identify the method that allows the friends to *most* effectively receive each other's messages, and explain how this method best meets the criteria *and* constraints. [1]

Method: Method 1/1/Morse code with flashlight/Morse code/flashlight

Explanation:

It can be seen by friends at all times, but the drum can only be used before 7:00 p.m.

OR

The flashlight that uses Morse code is cheap, can be used day and night, and is small and easily stored.

OR other valid response

ELS Training 1

17

Identify the method that allows the friends to *most* effectively receive each other's messages, and explain how this method best meets the criteria *and* constraints. [1]

Method:

Morse Code

Explanation:

I would say that morse code is the best if you want the form of communication to be quiet, not take up to much space, can be used day or night, and not very expensive. Morse code is all of these things. And it is not very hard to use or learn.

Score Credit 1 (out of 1 credit)

This response demonstrates a clear understanding of the concepts in the task.

- Morse code is chosen for the method.
- An acceptable explanation including how the chosen method best meets the criteria and constraints is provided.

This response is complete and correct.

ELS Training 2

17

Identify the method that allows the friends to *most* effectively receive each other's messages, and explain how this method best meets the criteria *and* constraints. [1]

Method:

Explanation:

Score Credit 1 (out of 1 credit)

This response demonstrates a fair understanding of the concepts in the task.

- Morse code with flashlight is chosen for the method.
- An acceptable explanation including how drum signals does not meet the criteria and constraints is provided.

This response is sufficient for credit.

ELS Training 3

17

Identify the method that allows the friends to *most* effectively receive each other's messages, and explain how this method best meets the criteria *and* constraints. [1]

Method: Morse code and flashlight

Explanation: This method is best meets of criteria and constraints because it is not loud after 7:00 p.m and it transfer easily from one to another.

Score Credit 1 (out of 1 credit)

This response demonstrates a fair understanding of the concepts in the task.

- Morse code and flashlight is chosen for the method.
- A minimally acceptable explanation including how the chosen method best meets the criteria and constraints is provided.

This response is sufficient for credit.

ELS Training 4

17

Identify the method that allows the friends to *most* effectively receive each other's messages, and explain how this method best meets the criteria *and* constraints. [1]

Method:

Explanation:

Score Credit 0 (out of 1 credit)

This response demonstrates an incomplete understanding of the concepts in the task.

- Drums and mallet is chosen for the method.
- An acceptable explanation including how drum signals does not meet the criteria and constraints is provided.

This response is insufficient for credit due to selection of the less effective method.

ELS Training 5

17

Identify the method that allows the friends to *most* effectively receive each other's messages, and explain how this method best meets the criteria *and* constraints. [1]

Method:

Explanation:

Score Credit 0 (out of 1 credit)

This response demonstrates an incomplete understanding of the concepts in the task.

- Morse code and flashlight is chosen for the method.
- The explanation does not specify the criteria and constraints that the use of drum signals does not meet.

This response is insufficient for credit due to an incomplete explanation.

ELS Training 6

17

Identify the method that allows the friends to *most* effectively receive each other's messages, and explain how this method best meets the criteria *and* constraints. [1]

Method:

morse code

Explanation:

Morse code meets the criteria the best because it follows all the criterias rules.

Score Credit 0 (out of 1 credit)

This response demonstrates an incomplete understanding of the concepts in the task.

- Morse code is chosen for the method.
- The explanation does not specify how the method best meets the criteria and constraints.

This response is insufficient for credit due to an incomplete explanation.

EXEMPLARY RESPONSE

18

Compare the amplitude and wavelength of Wave A to the amplitude and wavelength of Wave B. Mention both waves in your responses. [1]

Amplitude:

Amplitude of Wave A is greater/more than the amplitude of Wave B.

OR

A has an amplitude that is larger/taller/longer than B.

OR other valid response

Wavelength:

Wavelength of Wave A is the same as the wavelength of Wave B.

OR

Both waves have the same wavelength.

OR other valid response

ELS Training 1

18

Compare the amplitude and wavelength of Wave *A* to the amplitude and wavelength of Wave *B*. Mention both waves in your responses. [1]

Amplitude:

The amplitude on wave B is much smaller since it is not as loud as Wave A. Wave A's amplitude is much taller since it is louder unlike wave B.

Wavelength:

Both waves A and B have the same wavelength since they were both in the same pitch.

Score Credit 1 (out of 1 credit)

This response demonstrates a clear understanding of the concepts in the task.

- The amplitudes of Waves A and B are correctly compared.
- The wavelengths of Waves A and B are correctly compared.

This response is complete and correct.

ELS Training 2

18

Compare the amplitude and wavelength of Wave *A* to the amplitude and wavelength of Wave *B*. Mention both waves in your responses. [1]

Amplitude:

wave b has a lower amplitude because it is quieter. sound wave a is louder and has a higher amplitude

Wavelength:

the sound waves are both the same length

Score Credit 1 (out of 1 credit)

This response demonstrates a fair understanding of the concepts in the task.

- The amplitudes of Waves A and B are correctly compared.
- The wavelengths of Waves A and B are minimally correctly compared.

This response is sufficient for credit.

ELS Training 3

18

Compare the amplitude and wavelength of Wave *A* to the amplitude and wavelength of Wave *B*. Mention both waves in your responses. [1]

Amplitude:

The amplitude of Wave A is longer then Wave B. But besides the length they are very similar. This proves that Wave A is louder then Wave B

Wavelength:

The wavelength of both are the identical. This shows that they spoke for the same amount of time.

Score Credit 1 (out of 1 credit)

This response demonstrates a fair understanding of the concepts in the task.

- The amplitudes of Waves A and B are correctly compared.
- The wavelengths of Waves A and B are correctly compared. The additional information about the amount of time does not contradict the correct information about the wavelength.

This response is sufficient for credit.

ELS Training 4

18

Compare the amplitude and wavelength of Wave A to the amplitude and wavelength of Wave B. Mention both waves in your responses. [1]

Amplitude:

Wave A is higher in altitude than wave B because your speaking louder than whispering.

Wavelength:

The wave lenght is the same because you saying the samething as the other wave.

Score Credit 0 (out of 1 credit)

This response demonstrates limited understanding of the concepts in the task.

- The comparison of Waves A and B incorrectly refers to altitude instead of amplitude.
- The wavelengths of Waves A and B are correctly compared. The additional information about what is being said does not contradict the correct information about the wavelength.

This response is insufficient for credit due to the use of an incorrect term when comparing amplitudes.

ELS Training 5

18

Compare the amplitude and wavelength of Wave *A* to the amplitude and wavelength of Wave *B*. Mention both waves in your responses. [1]

Amplitude:

The amplitudes are different in "Sound wave A" and in "Sound wave B." To prove this, sound wave A is taller, and more deep. Sound wave B however is smaller, and is not as deep.

Wavelength:

The wavelength is a little bit different in both of them. The are both the same lighth horizontally, by it is not the same going up. One is taller.

Score Credit 0 (out of 1 credit)

This response demonstrates incomplete understanding of the concepts in the task.

- The amplitudes of Waves A and B are correctly compared.
- The wavelengths of Waves A and B are incorrectly compared as being different.

This response is insufficient for credit due to an incorrect wavelength comparison.

ELS Training 6

18

Compare the amplitude and wavelength of Wave *A* to the amplitude and wavelength of Wave *B*. Mention both waves in your responses. [1]

Amplitude:

The amplitude of wave A is more intense than wave B's, and therefore, makes a louder noise.

Wavelength:

The wavelength of wave A is the same as wave B, resulting in a similar noise.

Score Credit 0 (out of 1 credit)

This response demonstrates limited understanding of the concepts in the task.

- The comparison of Wave A being “more intense” than Wave B does not imply an increase in amplitude.
- The wavelengths of Waves A and B are correctly compared.

This response is insufficient for credit due to the use of an imprecise term when comparing amplitudes.

EXEMPLARY RESPONSE

25

Based on the information in the graphs, select **one** date from the table during which the greatest effects of Hurricane Harvey in Houston, Texas were recorded.

Date

August 27, 2017	
August 28, 2017	
August 29, 2017	

Then, complete the passage below by placing the correct term in *each* box from the choices given. Terms may be used more than once. [1]

Term Choices:

decrease

increase

remain constant

Passage

Based on the patterns seen in the graphs, Hurricane Harvey caused the air

temperature to decrease, the precipitation to increase,

and the wind speed to increase.

EXEMPLARY RESPONSE

28

Explain how the taste of the honey locust seed pods contributes to the reproduction of the honey locust trees. [1]

The sweet taste of the seed pods attracts animals to eat them and spread the seeds.

OR

Animals are attracted to the sweet taste of the seed pods. These seeds are eaten and eliminated in animal waste.

OR other valid response

ELS Training 1

28

Explain how the taste of the honey locust seed pods contributes to the reproduction of the honey locust trees. [1]

as the seed pods are sweet tasting, many animals may want to eat them. then, the animals poop them out somewhere else.

Score Credit 1 (out of 1 credit)

This response demonstrates a clear understanding of the concepts in the task.

- The explanation includes both attraction caused by the sweet taste and the resulting seed spread.

This response is complete and correct.

ELS Training 2

28

Explain how the taste of the honey locust seed pods contributes to the reproduction of the honey locust trees. [1]

The taste would contribute because more animals like to eat it meaning more manuer to spread the seeds.

Score Credit 1 (out of 1 credit)

This response demonstrates a fair understanding of the concepts in the task.

- The explanation includes both attraction caused by taste and the resulting seed spread.

This response is sufficient for credit.

ELS Training 3

28

Explain how the taste of the honey locust seed pods contributes to the reproduction of the honey locust trees. [1]

Because of there taste animal eat them then poop them out and planting them

Score Credit 1 (out of 1 credit)

This response demonstrates a fair understanding of the concepts in the task.

- The explanation implies attraction caused by taste and the resulting seed spread.

This response is sufficient for credit.

ELS Training 4

28

Explain how the taste of the honey locust seed pods contributes to the reproduction of the honey locust trees. [1]

The taste of the honey locust seed pods contrubutes to the reproduction of the honey locust trees, becouse they taste so good so the animals eat them but then theres nothing left to drop and grow.

Score Credit 0 (out of 1 credit)

This response demonstrates an incomplete understanding of the concepts in the task.

- The explanation includes attraction caused by taste. However, the explanation includes how reproduction is reduced, rather than supported.

This response is insufficient for credit because it does not address contribution to reproduction.

ELS Training 5

28

Explain how the taste of the honey locust seed pods contributes to the reproduction of the honey locust trees. [1]

The taste of honey locust seed pods contributes to the reproduction of the honey locust trees because they fall, animals eat it, and then it get planted again when they have droppings.

Score Credit 0 (out of 1 credit)

This response demonstrates an incomplete understanding of the concepts in the task.

- The explanation does not include attraction caused by taste but does include seed spread.

This response is insufficient for credit because it does not address why the seed pods are eaten.

ELS Training 6

28

Explain how the taste of the honey locust seed pods contributes to the reproduction of the honey locust trees. [1]

The taste of honey locust seed pod contributes because. Animals eat the honey from the tree that makes it reproduce seed pods,.

Score Credit 0 (out of 1 credit)

This response demonstrates an incomplete understanding of the concepts in the task.

- The explanation includes implied attraction caused by honey taste. However, the explanation does not adequately describe reproduction.

This response is insufficient for credit because it does not address contribution to reproduction.

EXEMPLARY RESPONSE

33

Offshore drilling can have negative effects on the environment. Describe **one negative** effect that may have led New York State to ban offshore drilling. [1]

Offshore drilling can cause oil spills on the ocean surface, which pollutes the water.

OR

Offshore drilling can cause noise pollution, which affects ocean animals and people living near the shore.

OR

Offshore drilling can interfere with the habitat of fish/birds.

OR

Oil spills can harm the animals/plants in the environment.

OR other valid response

ELS Training 1

33

Offshore drilling can have negative effects on the environment. Describe **one negative** effect that may have led New York State to ban offshore drilling. [1]

One negative effect that may have led New York to ban offshore drilling is that marine birds, wich inhabbit New York shores, may have been diving for fish, getting coated with oil, then dying.

Score Credit 1 (out of 1 credit)

This response demonstrates a clear understanding of the concepts in the task.

- An appropriate negative effect on the environment is provided.

This response is complete and correct.

ELS Training 2

33

Offshore drilling can have negative effects on the environment. Describe **one negative** effect that may have led New York State to ban offshore drilling. [1]

Offshore drilling can pollute the ocean

Score Credit 1 (out of 1 credit)

This response demonstrates a fair understanding of the concepts in the task.

- An adequate negative effect on the environment is provided.

This response is sufficient for credit.

ELS Training 3

33

Offshore drilling can have negative effects on the environment. Describe **one negative** effect that may have led New York State to ban offshore drilling. [1]

One negative effect offshore drilling could have is that it's destroying animals habitats.

Score Credit 1 (out of 1 credit)

This response demonstrates a fair understanding of the concepts in the task.

- An adequate negative effect on the environment is provided.

This response is sufficient for credit.

ELS Training 4

33

Offshore drilling can have negative effects on the environment. Describe **one negative** effect that may have led New York State to ban offshore drilling. [1]

One negative effect of offshore drilling is that it might affect and animals habbitat.

Score Credit 0 (out of 1 credit)

This response demonstrates an incomplete understanding of the concepts in the task.

- The effect provided is described so generally that it is not clear that it is specifically a negative effect.

This response is insufficient for credit due to an incomplete description of a negative effect.

ELS Training 5

33

Offshore drilling can have negative effects on the environment. Describe **one negative** effect that may have led New York State to ban offshore drilling. [1]

One negative effect is that it could take away the oil from underground.

Score Credit 0 (out of 1 credit)

This response demonstrates no overall understanding of the concepts in the task.

- The effect provided is not linked with a negative effect on the environment.

This response is insufficient for credit because it does not address negative environmental impact.

ELS Training 6

33

Offshore drilling can have negative effects on the environment. Describe **one negative** effect that may have led New York State to ban offshore drilling. [1]

One negative effect is oil seep. oil seep occurs when crude oil leaks through natural cracks in the ocean floor.

Score Credit 0 (out of 1 credit)

This response demonstrates no overall understanding of the concepts in the task.

- The effect provided is not linked to an associated negative effect on the environment.

This response is insufficient for credit because it does not address negative environmental impact.

EXEMPLARY RESPONSE

34

Describe how the Maple Ridge Wind Farm helps protect the environment. [1]

Wind turbines do not produce air pollutants.

OR

Wind turbines do not disturb animal habitats significantly.

OR

Wind turbines produce electricity without burning fossil fuels, which pollute the environment.

OR other valid response

ELS Training 1

34

Describe how the Maple Ridge Wind Farm helps protect the environment. [1]

Maple Ridge Wind Farm helps protect the environment. If there were no wind mills there might be oil diggers instead. The animals would lose their homes and there will be less oil over all.

Score Credit 1 (out of 1 credit)

This response demonstrates a fair understanding of the concepts in the task.

- The description that animal habitats would not be lost adequately explains how the environment is protected.

This response is sufficient for credit.

ELS Training 2

34

Describe how the Maple Ridge Wind Farm helps protect the environment. [1]

Maple Ridge Farm is helping the environment. For example it is not poluting the earth as much.

Score Credit 1 (out of 1 credit)

This response demonstrates a fair understanding of the concepts in the task.

- The description of less pollution adequately explains how the environment is protected.

This response is sufficient for credit.

ELS Training 3

34

Describe how the Maple Ridge Wind Farm helps protect the environment. [1]

The Maple Ridge wind farm helps protect the environment because the wind turbines aren't drilling into the sea floor or using excavators to mine for coal.

Score Credit 1 (out of 1 credit)

This response demonstrates a fair understanding of the concepts in the task.

- The description of less harm to Earth's surface adequately explains how the environment is protected.

This response is sufficient for credit.

ELS Training 4

34

Describe how the Maple Ridge Wind Farm helps protect the environment. [1]

Maple Ridge Wind Farm helps protect the environment becaues they use wind insted on burning coal and having global warming.

Score Credit 1 (out of 1 credit)

This response demonstrates a fair understanding of the concepts in the task.

- The description of less global warming adequately explains how the environment is protected.

This response is sufficient for credit.

ELS Training 5

34

Describe how the Maple Ridge Wind Farm helps protect the environment. [1]

the Maple Ridge Wind Farm helps protect the environment by makeing more electricity with large windmills called wind turbines.

Score Credit 0 (out of 1 credit)

This response demonstrates no overall understanding of the concepts in the task.

- The description provided describes the function of windmills and not how they protect the environment.

This response is insufficient for credit due to a description that does not answer the question.

ELS Training 6

34

Describe how the Maple Ridge Wind Farm helps protect the environment. [1]

The Maple Ridge Wind Farm helps the environment by using wind to provide energy for electricity which can help conserve Earth's natural resources.

Score Credit 0 (out of 1 credit)

This response demonstrates incomplete understanding of the concepts in the task.

- The description provided is too general and does not specify how the environment is protected.

This response is insufficient for credit due to a description that does not fully answer the question.

ELS Training 7

34

Describe how the Maple Ridge Wind Farm helps protect the environment. [1]

The Maple Ridge Wind Farm heps protect the envrioment by not damaging things.

Score Credit 0 (out of 1 credit)

This response demonstrates incomplete understanding of the concepts in the task.

- The description provided is too general and does not specify how the environment is protected.

This response is insufficient for credit due to a description that does not fully answer the question.